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Project Title: Uncovering Deceptive Reviews Using Machine Learning and NLP Techniques with Power BI for Strategic Insights

Submitted in partial fulfillment of the requirements for the award of the degree of

Master of Science (Data Science) MSc (DS)

By

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Enrollment No: O24MSD110026

Under the guidance of

Guide/Mentor Name: Kunwar Saurabh Bisen

Certificate

Copy to the certificate received from Qollabb to be pasted here.

Declaration

I, **Tushar Goswami**, hereby solemnly declare that the project report titled " **Uncovering Deceptive Reviews Using Machine Learning and NLP Techniques with Power BI for Strategic Insights** " submitted in partial fulfillment of the requirements for the award of the degree of **Master of Science (Data Science) MSc (DS)** is my original work.

I further declare that:

- This project has been carried out by me during the academic year **2026** under the supervision of **Kunwar Saurabh Bisen (Guide/Mentor Name)**.
- The work has not been submitted previously to any other university, institution, or examination body for the award of any degree, diploma, or certification.
- All sources of information used in this report have been duly acknowledged and referenced in accordance with academic ethics and plagiarism norms.
- The data presented in this report is authentic to the best of my knowledge and has not been fabricated or manipulated.
- I understand that if any part of this declaration is found to be false, my project may be rejected and disciplinary action may be taken as per institutional rules.

Place: Vrindavan

Date: 04-07-26

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Acknowledgement

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“Uncovering Deceptive Reviews Using Machine Learning and NLP Techniques with Power BI for Strategic Insights”

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ML	Machine Learning
NLP	Natural Language Processing
TF-IDF	Term Frequency – Inverse Document Frequency
NLTK	Natural Language Toolkit
API	Application Programming Interface
DFD	Data Flow Diagram
UML	Unified Modeling Language
ER	Entity Relationship
KPI	Key Performance Indicator
CSV	Comma Separated Values
UI	User Interface
UX	User Experience
SVM	Support Vector Machine
VADER	Valence Aware Dictionary and Sentiment Reasoner
AWS	Amazon Web Services

Abstract

Online reviews have become one of the most influential factors affecting students' decisions while selecting educational institutions. Students frequently depend on online ratings and feedback available on various educational platforms to evaluate important aspects such as faculty quality, placement opportunities, infrastructure, course curriculum, and overall student experience. However, the increasing presence of deceptive or manipulated reviews has created concerns regarding the authenticity and reliability of such information. Fake reviews can mislead prospective students and negatively impact trust in online educational review systems.

This project focuses on detecting deceptive reviews in the education sector using Machine Learning and Natural Language Processing (NLP) techniques. One of the primary challenges faced during the project was the limited availability of labeled fake review datasets specifically related to educational institutions. To overcome this issue, a cross-domain learning approach was adopted by utilizing publicly available fake review datasets from platforms such as Kaggle, Yelp, and HuggingFace, while education-related reviews were collected through web scraping from Collegedunia, Career360, and Google Reviews.

The collected review data was preprocessed using NLP techniques such as tokenization, stopword removal, text normalization, and TF-IDF feature extraction. A Logistic Regression classifier was trained to classify reviews as fake or genuine. In addition, sentiment analysis was performed using the VADER sentiment analyzer from the NLTK library to categorize reviews into positive, neutral, and negative sentiments.

The final results were visualized using Power BI dashboards containing KPIs, sentiment distribution charts, fake review analysis, and institutional insights. Out of 356 reviews analyzed, approximately 5.3% were identified as potentially deceptive. The project demonstrates how machine learning, NLP, and data visualization techniques can be combined to improve the reliability of online review systems in the education sector.

Chapter 1: Introduction

1.1 Background of the Study

In the digital era, online reviews have become an essential source of information for individuals making important decisions. In the education sector, students increasingly depend on online ratings and reviews to evaluate colleges, universities, courses, faculty quality, placements, infrastructure, and overall student experience. Educational review platforms such as Collegedunia, Career360, and Google Reviews provide students with access to large volumes of feedback shared by current and former students.

The influence of online reviews is significant because students often use these reviews to compare institutions and make career-related decisions. Positive reviews can improve the reputation of educational institutions, while negative reviews can affect public perception and student enrollment. As a result, online reviews have become a powerful factor in shaping institutional image and credibility.

However, the growing popularity of online review systems has also increased the risk of deceptive or manipulated reviews. Fake reviews are intentionally created to promote or damage the reputation of institutions by providing misleading information. These reviews may be generated by institutions themselves, third-party marketing agencies, or individuals attempting to manipulate public perception. Such misleading reviews reduce the reliability of online feedback systems and create difficulties for students attempting to make informed decisions.

The detection of fake reviews has been widely studied in domains such as e-commerce, hotels, restaurants, and product review systems. However, comparatively less research has been conducted in the education sector. This creates a research gap and highlights the need for systems capable of identifying deceptive reviews related to educational institutions.

Machine Learning (ML) and Natural Language Processing (NLP) techniques provide effective solutions for detecting deceptive textual patterns in online reviews. By analyzing review content, writing patterns, sentiment, and linguistic features, ML models can classify reviews as genuine or fake. NLP techniques further help in understanding sentiment polarity and extracting meaningful textual insights.

This project aims to apply ML and NLP techniques to identify deceptive reviews in educational institutions and visualize insights using Power BI dashboards. The project

combines fake review datasets from different industries with scraped education-related reviews to develop a practical fake review detection system.

1.2 Problem Statement

The rapid growth of online educational review platforms has made reviews an important factor influencing students' academic decisions. However, the presence of fake or manipulated reviews reduces the credibility and reliability of these platforms. Deceptive reviews can mislead students by presenting inaccurate or exaggerated information regarding educational institutions.

One of the major challenges in developing fake review detection systems for the education sector is the lack of publicly available labeled datasets specifically focused on educational reviews. Most existing fake review datasets are related to domains such as hotels, restaurants, and e-commerce products. As a result, directly training machine learning models for educational reviews becomes difficult.

Another challenge is that fake reviews often resemble genuine reviews in writing style, making manual detection difficult. Traditional moderation systems may not effectively identify suspicious patterns within large volumes of textual data.

Therefore, there is a need for an automated system capable of analyzing educational reviews using Machine Learning and NLP techniques to identify potentially deceptive reviews and provide meaningful insights regarding review authenticity.

1.3 Objectives of the Project

The primary objectives of this project are as follows:

1. To develop a Machine Learning model capable of detecting deceptive reviews in the education sector.
2. To apply Natural Language Processing techniques for analyzing textual patterns and sentiments within reviews.
3. To preprocess and clean review datasets collected from multiple online sources.
4. To perform sentiment analysis on educational reviews using NLP techniques.
5. To create interactive Power BI dashboards for visualizing fake review patterns, sentiment distribution, and institutional insights.

6. To generate strategic insights and recommendations based on review authenticity analysis.
-

1.4 Scope of the Project

The scope of this project is focused on detecting deceptive reviews related to educational institutions using Machine Learning and NLP techniques. The project involves collecting reviews from publicly available educational review platforms and analyzing them using text processing and classification methods.

The scope includes:

- Collection of fake review datasets from Kaggle, Yelp, and HuggingFace.
- Scraping educational reviews from Collegedunia, Career360, and Google Reviews.
- Preprocessing and cleaning textual review data.
- Feature extraction using TF-IDF vectorization.
- Sentiment analysis using the VADER sentiment analyzer.
- Fake review prediction using Logistic Regression.
- Visualization of insights using Power BI dashboards.

The project is limited to textual review analysis and does not include multimedia content such as images or videos. The analysis is also limited by the size and availability of educational review datasets.

1.5 Existing System

Existing online educational review systems mainly rely on user-generated reviews and ratings without strong mechanisms for identifying deceptive content. Most review platforms display reviews publicly without advanced authenticity verification methods. As a result, fake reviews may remain undetected and influence users' perceptions.

Traditional moderation systems primarily depend on manual review verification or basic spam filtering techniques. These approaches are often insufficient when dealing with large-scale review datasets. Additionally, most existing fake review detection research focuses on e-commerce and hospitality domains rather than educational institutions.

The absence of domain-specific fake review detection systems in the education sector highlights the limitations of current systems and the need for more advanced analytical approaches.

1.6 Proposed System

The proposed system aims to develop an intelligent fake review detection framework using Machine Learning and Natural Language Processing techniques.

The proposed system performs the following tasks:

- Collection and preprocessing of review data.
- Text cleaning and normalization.
- Feature extraction using TF-IDF vectorization.
- Fake review classification using Logistic Regression.
- Sentiment analysis using VADER.
- Visualization of analytical insights using Power BI dashboards.

The system is designed to automatically identify potentially deceptive reviews and provide users with insights regarding review authenticity and sentiment patterns.

Compared to traditional systems, the proposed approach improves scalability, automation, and analytical capabilities by utilizing data-driven techniques for review analysis.

1.7 Technologies Used

The successful implementation of this project required the integration of multiple technologies, programming libraries, and analytical tools. These technologies were selected based on their capability to support data preprocessing, machine learning, natural language processing, visualization, and dashboard development.

Python

Python was used as the primary programming language for the project due to its simplicity, flexibility, and extensive support for Machine Learning and NLP libraries. Python enabled efficient implementation of data preprocessing, model training, sentiment analysis, and data manipulation tasks.

Pandas

The Pandas library was used for data manipulation and preprocessing. It helped in handling tabular datasets, cleaning review data, removing duplicates, filtering educational reviews, and performing data transformations efficiently.

NumPy

NumPy was used for numerical computations and array operations during data preprocessing and machine learning implementation. It provided efficient mathematical operations required for feature handling and model processing.

Scikit-learn

Scikit-learn was used for Machine Learning model development. It provided tools for TF-IDF vectorization, train-test splitting, Logistic Regression classification, and model evaluation. The library helped implement the fake review classification system efficiently.

Natural Language Toolkit (NLTK)

NLTK was used for Natural Language Processing tasks such as tokenization, stopword removal, and sentiment analysis. The VADER sentiment analyzer available in NLTK was used to classify reviews into positive, negative, and neutral sentiments.

TF-IDF Vectorization

TF-IDF vectorization was selected because machine learning algorithms cannot directly process raw textual information. The technique assigns numerical weights to words based on their frequency within a document and across the complete dataset. Frequently occurring words that appear in almost every review receive lower importance, whereas unique and informative words receive higher weights. This representation enables the Logistic Regression classifier to distinguish between genuine and deceptive reviews by learning linguistic patterns associated with educational feedback. Compared with simple Bag-of-Words models, TF-IDF reduces the influence of common terms while emphasizing discriminative vocabulary, resulting in improved text representation and better classification performance.

Power BI

Power BI was used to create interactive dashboards and visualizations for the final analysis. Various charts, KPIs, sentiment distributions, and fake review insights were presented through Power BI dashboards to improve interpretability and decision-making.

Jupyter Notebook

Jupyter Notebook was used as the development environment for implementing Python code, data preprocessing, machine learning workflows, and NLP analysis. It provided an interactive environment for experimentation and analysis.

Web Scraping Tools

Educational reviews were collected from publicly available platforms such as Collegedunia, Career360, and Google Reviews using web scraping techniques. The scraped data was then processed and analyzed within the project workflow.

Chapter 2: System Study

2.1 Introduction

Online reviews have become one of the most influential factors affecting customer decisions across various industries. Whether purchasing a product, booking a hotel, selecting a healthcare provider, or choosing an educational institution, users increasingly rely on reviews to evaluate the quality and credibility of services. Genuine reviews help prospective customers make informed decisions by providing authentic experiences shared by previous users. However, the growing dependence on online reviews has also led to the emergence of deceptive or fake reviews, which are intentionally created to manipulate public perception.

Fake reviews are written with the objective of either promoting or damaging the reputation of an organization, product, or service. These reviews are often generated by individuals hired for promotional purposes, competitors attempting to reduce an organization's credibility, or automated systems capable of generating human-like textual content. Such deceptive information misleads users and negatively affects trust in online platforms.

The education sector has experienced significant digital transformation in recent years. Students frequently consult online review platforms before selecting colleges, universities, or training institutions. Websites such as Collegedunia, Career360, and Google Reviews provide thousands of student reviews discussing academic quality, faculty, infrastructure, placements, hostel facilities, and overall campus experience. Although these reviews assist students in making educational decisions, they are also vulnerable to manipulation through fake or biased reviews.

Unlike e-commerce and hospitality sectors, publicly available datasets specifically containing fake educational reviews are extremely limited. Most existing fake review datasets are collected from hotel booking platforms, product review websites, and restaurant applications. Consequently, researchers face considerable challenges while developing machine learning models dedicated to educational review analysis.

To overcome this limitation, the present study combines publicly available fake review datasets with real educational reviews collected through web scraping. A filtering mechanism based on education-specific keywords is implemented to retain only those reviews relevant to educational institutions before training the machine learning model. This domain adaptation approach enables the development of a fake review detection system despite the absence of a dedicated educational fake review dataset.

The proposed system integrates Natural Language Processing (NLP), Machine Learning (ML), and Business Intelligence techniques to identify deceptive reviews and generate actionable insights through interactive Power BI dashboards. The combination of these technologies enables educational institutions and prospective students to better understand review authenticity while supporting data-driven decision making.

2.2 Importance of Online Reviews in the Education Sector

In today's digital era, online reviews have become an essential source of information for students seeking admission to colleges and universities. Before applying to an institution, students often explore online platforms to understand academic quality, placement opportunities, faculty expertise, campus infrastructure, hostel facilities, extracurricular activities, and student satisfaction. These reviews significantly influence institutional reputation and student enrollment decisions.

Educational review platforms provide transparency by allowing current and former students to share their personal experiences. Positive reviews can enhance an institution's credibility and attract prospective applicants, whereas negative reviews may discourage admissions and impact institutional rankings. As a result, maintaining trustworthy reviews has become increasingly important for educational organizations.

However, the popularity of online educational reviews has also introduced the risk of review manipulation. Some institutions or third parties may intentionally publish exaggerated positive reviews to improve institutional image, while competitors may post misleading negative reviews to damage reputation. Such deceptive practices reduce the reliability of online review platforms and make it difficult for students to distinguish between genuine experiences and fabricated opinions.

Unlike product reviews, educational reviews often contain detailed descriptions related to faculty behavior, curriculum design, placement support, classroom environment, research opportunities, and long-term career outcomes. These domain-specific characteristics make fake review detection more challenging because educational experiences are subjective and vary considerably among students.

The increasing dependence on online educational reviews highlights the necessity of intelligent systems capable of identifying suspicious review patterns. Machine learning algorithms combined with natural language processing techniques can automatically analyze textual content and classify reviews as genuine or deceptive based on linguistic characteristics, writing style, sentiment, and contextual information.

Therefore, fake review detection has become an important research area within educational data analytics. Developing reliable systems for identifying deceptive educational reviews not only improves transparency but also assists students, parents, educational institutions, and policymakers in making informed decisions based on trustworthy information.

2.3 Review of Similar Systems

Several systems and research studies have been developed to identify fake reviews across domains such as e-commerce, hospitality, restaurant services, and product reviews. Most existing systems focus on detecting spam reviews using textual analysis, reviewer behavior, sentiment analysis, and machine learning classification techniques.

2.3.1 E-Commerce Review Detection Systems

E-commerce platforms such as Amazon and Flipkart rely heavily on user reviews to influence customer purchasing decisions. Due to the financial impact of positive ratings, fake reviews have become common in e-commerce platforms.

Many existing fake review detection systems use supervised machine learning algorithms such as:

- Logistic Regression
- Support Vector Machine (SVM)
- Naive Bayes
- Random Forest

These models analyze textual features such as word frequency, review length, sentiment polarity, and writing patterns to classify reviews as fake or genuine.

Although these systems perform effectively in product review environments, they are not specifically designed for educational reviews, where writing style and review characteristics may differ significantly.

2.3.2 Hotel and Restaurant Review Detection Systems

Fake review detection has also been widely studied in hospitality platforms such as Yelp and TripAdvisor. In these systems, NLP techniques are used to identify suspicious

linguistic patterns, exaggerated sentiments, and repetitive phrases commonly associated with fake reviews.

Some hospitality review systems use behavioral analysis in addition to textual analysis. These systems monitor:

- Reviewer activity patterns
- Posting frequency
- Rating distributions
- Duplicate content

While such systems are effective for hospitality domains, educational review platforms usually have fewer user interactions and less reviewer metadata, making direct implementation difficult.

2.3.3 Educational Review Platforms

Educational review platforms primarily focus on displaying student opinions regarding institutions, faculty, placements, and campus facilities. However, most platforms currently rely on manual moderation systems or basic spam filtering techniques.

Unlike e-commerce platforms, educational review systems rarely implement advanced machine learning-based fake review detection mechanisms. This creates opportunities for misleading reviews to influence institutional reputation and student decision-making.

The absence of dedicated fake review detection systems in educational platforms highlights the need for domain-specific analytical approaches.

2.4 Comparative Analysis of Existing Systems

Different fake review detection systems use varying approaches depending on the domain and available data. The following table summarizes the comparison between existing systems and the proposed system.

Table 2.1

Feature	E-Commerce Systems	Hospitality Systems	Proposed Education System
Domain	Products	Hotels & Restaurants	Educational Institutions
Data Type	Product Reviews	Service Reviews	Educational Reviews
Detection Method	ML Classification	NLP + Behavioral Analysis	ML + NLP
Sentiment Analysis	Limited	Extensive	Extensive
Dashboard Visualization	Rare	Moderate	Power BI Dashboard
Domain Adaptation	Not Required	Not Required	Required
Educational Focus	No	No	Yes

The proposed system differs from traditional systems because it specifically focuses on educational reviews and incorporates domain adaptation techniques to align cross-domain datasets with the education sector.

2.5 Machine Learning Approaches Used in Fake Review Detection

Machine Learning plays a major role in identifying deceptive reviews by learning textual patterns from labeled datasets.

Logistic Regression

Logistic Regression is one of the most commonly used supervised learning algorithms for text classification problems. It is computationally efficient and performs well for binary classification tasks such as identifying fake and genuine reviews.

In this project, Logistic Regression was selected because:

- It performs effectively on TF-IDF features.
 - It is suitable for smaller datasets.
 - It provides interpretable classification results.
-

Support Vector Machine (SVM)

Support Vector Machine is another widely used classification algorithm in fake review detection systems. SVM attempts to separate fake and genuine reviews by finding an optimal decision boundary.

Although SVM provides strong performance in many text classification problems, Logistic Regression was preferred in this project due to its simplicity and faster training process.

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification accuracy. It is effective for structured datasets but may require higher computational resources for textual feature analysis.

Naive Bayes

Naive Bayes classifiers are commonly used for spam filtering and sentiment analysis due to their probabilistic approach. However, they assume feature independence, which may reduce performance when analyzing complex textual relationships.

2.6 Natural Language Processing Techniques

Natural Language Processing (NLP) techniques are essential for analyzing textual review data and extracting meaningful information.

Text Preprocessing

Text preprocessing is the first stage of NLP analysis. Raw review text often contains noise such as punctuation, numbers, special characters, and irrelevant words.

The preprocessing steps used in this project include:

- Lowercase conversion
- Tokenization
- Stopword removal
- Text normalization

- Duplicate removal

These preprocessing techniques improve the quality and consistency of textual data before machine learning analysis.

TF-IDF Vectorization

TF-IDF (Term Frequency–Inverse Document Frequency) is one of the most widely used feature extraction techniques in NLP.

TF-IDF converts textual reviews into numerical vectors by calculating:

- Frequency of words within a review
- Importance of words across the dataset

This helps Machine Learning models identify important textual patterns and distinguish between fake and genuine reviews.

Sentiment Analysis

Sentiment analysis is used to determine the emotional tone of textual reviews. In this project, sentiment analysis was performed using the VADER sentiment analyzer from the NLTK library.

VADER assigns sentiment scores and classifies reviews into:

- Positive
- Neutral
- Negative

Sentiment analysis helps identify exaggerated or suspicious emotional patterns that may indicate deceptive reviews.

2.7 Software Development Models

Software development models provide structured approaches for designing and implementing systems.

Waterfall Model

The Waterfall model follows a sequential development process consisting of:

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Deployment

The Waterfall approach is simple and easy to manage but lacks flexibility when project requirements change frequently.

Agile Model

The Agile model focuses on iterative and incremental development. Agile allows continuous improvements and flexibility throughout the project lifecycle.

In this project, an Agile-inspired approach was followed because:

- Data preprocessing required multiple iterations.
- Model tuning and testing were performed repeatedly.
- Dashboard visualizations evolved during analysis.

The iterative nature of Agile development helped improve project adaptability and experimentation.

2.8 Relevant Frameworks and Libraries

Several frameworks and libraries were used in the development of the proposed system.

Pandas

Pandas was used for data cleaning, preprocessing, filtering, and tabular data manipulation.

NumPy

NumPy supported numerical computations and matrix operations required for machine learning workflows.

Scikit-learn

Scikit-learn provided machine learning tools for:

- TF-IDF vectorization
 - Logistic Regression
 - Model training
 - Train-test splitting
 - Evaluation metrics
-

NLTK

NLTK was used for:

- Stopword removal
 - Tokenization
 - Sentiment analysis
 - Text normalization
-

Power BI

Power BI was used for creating interactive dashboards and visual analytics for fake review analysis.

2.9 Research Gap

Although significant research has been conducted on fake review detection, several limitations still exist in current systems.

Most existing research focuses on domains such as:

- E-commerce
- Hospitality
- Product reviews

Very limited work has been performed specifically for educational review systems. Additionally, many studies rely heavily on reviewer behavioral metadata, which may not be available in educational platforms.

Another major gap is the lack of publicly available labeled datasets for educational fake reviews. This creates challenges for developing domain-specific machine learning models.

The proposed project attempts to address these limitations by:

- Applying domain adaptation techniques
- Combining cross-domain fake review datasets
- Scraping education-related reviews
- Integrating ML, NLP, and Power BI visualization

This approach helps create a more practical and domain-focused fake review detection system for educational institutions.

2.10 Characteristics of Educational Reviews

Educational reviews differ significantly from reviews found in domains such as e-commerce and hospitality. Students often discuss multiple aspects of an institution within a single review, including faculty, placements, campus infrastructure, academic curriculum, laboratory facilities, hostel accommodation, examination patterns, and student life. Consequently, educational reviews tend to be longer, more descriptive, and more opinion-oriented than product reviews.

The following examples illustrate the diversity of reviews collected during this study.

Example 1: Placement-Oriented Review

"Placements at DTU are generally very good, with many reputed companies visiting the campus every year. Students receive opportunities in tech, consulting, and core sectors with competitive salary packages."

This review primarily discusses placement opportunities and industry exposure. The language is balanced and contains institution-specific details.

Example 2: Negative Student Experience

"I never seen in my life this much of worst college... they torture the students and parents... management will collect money from students."

This review expresses strong dissatisfaction and contains highly emotional language. Reviews with extreme wording require careful analysis because they may represent either genuine frustration or exaggerated opinions.

Example 3: Mixed Opinion Review

"Some teachers are good and some are bad... placement and training cell are not good... I personally recommend trying another college."

This review contains both positive and negative observations, making it an example of mixed sentiment. Such reviews demonstrate why sentiment analysis alone is insufficient for fake review detection.

Example 4: Balanced Review

"The faculty to student ratio is good... teachers are approachable... labs are well maintained... exams are easy if students study."

This review provides detailed observations about multiple aspects of the institution and represents the type of structured feedback commonly found in educational review platforms.

These examples demonstrate that educational reviews frequently contain multiple topics within a single piece of text, including academics, infrastructure, faculty behaviour, placement opportunities, student life, and administrative policies. Unlike short product reviews, educational reviews often express complex opinions and mixed sentiments. Therefore, effective fake review detection requires not only sentiment analysis but also machine learning techniques capable of identifying linguistic patterns and contextual features associated with deceptive content.

2.11 Challenges in Collecting Educational Review Data

One of the major challenges encountered during this project was the absence of publicly available labeled fake review datasets specifically for educational institutions. Existing benchmark datasets from Kaggle and Hugging Face primarily contain reviews from domains such as hotels, restaurants, movies, electronics, and consumer products. Directly using these datasets would introduce a significant domain mismatch because the vocabulary, writing style, and review context differ substantially from educational reviews.

To address this limitation, a domain adaptation strategy was adopted. Education-related reviews were extracted from the benchmark dataset using keyword filtering based on terms such as *college*, *university*, *faculty*, *placement*, *students*, *semester*, and *course*. Additionally, 356 real educational reviews were scraped from Collegedunia, Career360, and Google Reviews. The filtered benchmark dataset was then used to train the machine learning model, while the scraped educational reviews were used to evaluate the model and generate analytical insights through Power BI.

2.12 Summary

This chapter reviewed existing fake review detection systems, machine learning approaches, NLP techniques, development methodologies, and relevant frameworks used in review analysis systems.

The analysis revealed that although fake review detection has been widely studied in domains such as e-commerce and hospitality, limited research exists for educational institutions. Existing systems also lack education-specific datasets and analytical visualization components.

The proposed system addresses these limitations by integrating Machine Learning, NLP techniques, sentiment analysis, and Power BI dashboards specifically for the education sector.

• Operational feasibility • System Architecture • DFD (Level 0, Level 1)

Chapter 3: System Analysis

3.1 Introduction

System analysis is an important phase in software development that focuses on understanding system requirements, identifying problems, analyzing user needs, and designing an efficient solution. The purpose of system analysis is to ensure that the proposed system fulfills functional and operational requirements effectively.

In this project, system analysis was performed to understand the challenges associated with fake reviews in educational platforms and to design a Machine Learning and NLP-based solution capable of detecting deceptive reviews. The analysis phase includes functional requirements, non-functional requirements, user requirements, feasibility studies, system architecture, and data flow analysis.

The proposed system is designed to automate the process of fake review detection and provide analytical insights through interactive dashboards.

3.2 Functional Requirements

Functional requirements describe the specific operations and functionalities that the system must perform.

The proposed fake review detection system includes the following functional requirements:

3.2.1 Data Collection

The system must collect educational review data from publicly available platforms such as:

- Collegedunia
- Career360
- Google Reviews

The collected data should include:

- Institution name
- Review text

- Ratings
 - Source platform
-

3.2.2 Data Preprocessing

The system must preprocess the collected review data by:

- Converting text to lowercase
- Removing punctuation and special characters
- Removing stopwords
- Removing duplicate reviews
- Cleaning noisy textual data

The preprocessing module ensures that the dataset is suitable for machine learning analysis.

3.2.3 Feature Extraction

The system must convert textual review data into numerical representations using TF-IDF vectorization.

This functionality allows Machine Learning models to analyze textual patterns mathematically.

3.2.4 Fake Review Detection

The system must classify reviews as:

- Fake
- Genuine

using a supervised Machine Learning model based on Logistic Regression.

3.2.5 Sentiment Analysis

The system must analyze review sentiment and classify reviews into:

- Positive
- Neutral
- Negative

using the VADER sentiment analyzer.

3.2.6 Dashboard Visualization

The system must provide visual insights using Power BI dashboards including:

- Fake vs Genuine review distribution
 - Sentiment distribution
 - Fake reviews by institution
 - Rating vs sentiment analysis
 - KPI metrics
-

3.2.7 Data Export

The system should allow processed datasets and prediction outputs to be exported for reporting and further analysis.

3.3 Non-Functional Requirements

Non-functional requirements describe system quality attributes and operational characteristics.

3.3.1 Performance

The system should process review datasets efficiently and generate predictions within acceptable response time.

3.3.2 Scalability

The system should support future expansion with larger datasets and additional review sources.

3.3.3 Accuracy

The Machine Learning model should provide reliable fake review predictions with acceptable classification accuracy.

3.3.4 Usability

The Power BI dashboard should be user-friendly and easy to interpret for both technical and non-technical users.

3.3.5 Maintainability

The system should allow future updates such as:

- Adding new machine learning models
 - Integrating deep learning techniques
 - Including new review platforms
-

3.3.6 Security

The system should ensure safe handling of scraped data and avoid unauthorized modification of datasets.

3.4 User Requirements

The proposed system is designed for users such as:

- Students
- Educational analysts
- Researchers

- Educational review platforms

The user requirements include:

- Ability to identify fake reviews
- Access to sentiment analysis insights
- Visualization of review authenticity
- Institution-level review analysis
- Interactive filtering and exploration of data

Users should be able to interpret dashboard insights easily without requiring advanced technical knowledge.

3.5 Feasibility Study

Feasibility analysis evaluates whether the proposed system is practical and implementable.

3.5.1 Technical Feasibility

The project is technically feasible because all required technologies, frameworks, and tools are readily available.

The project uses:

- Python
- Scikit-learn
- NLTK
- Power BI
- Pandas
- NumPy

The required software tools are open-source and supported by extensive documentation and community resources.

The hardware requirements are also moderate, making the system suitable for development on standard personal computers.

3.5.2 Economic Feasibility

The project is economically feasible because most technologies and libraries used are free and open-source.

No expensive infrastructure or licensing costs were required for:

- Python
- Scikit-learn
- NLTK
- Jupyter Notebook

Power BI Desktop was used for dashboard development, which is also available free for academic purposes.

Therefore, the overall implementation cost of the project remains low.

3.5.3 Operational Feasibility

The system is operationally feasible because it simplifies the process of analyzing educational reviews and detecting deceptive content automatically.

The Power BI dashboard provides an interactive interface for users to explore review insights without requiring advanced programming knowledge.

The system can assist educational platforms and students in identifying suspicious review patterns and improving trust in online reviews.

3.6 System Architecture

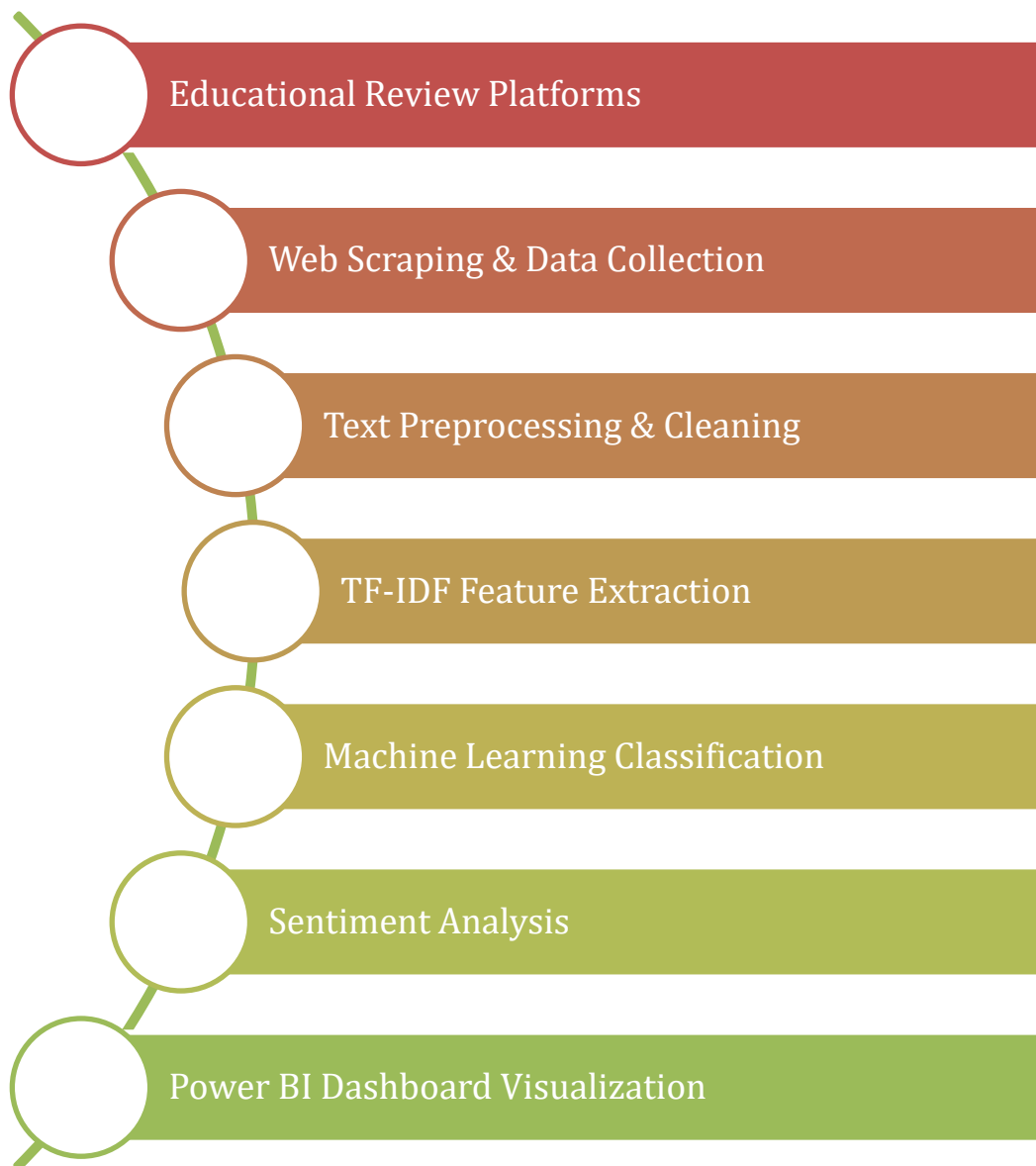
The proposed system follows a structured workflow consisting of multiple stages.

The major components of the system are:

1. Data Collection Layer
2. Data Preprocessing Layer
3. Feature Extraction Layer
4. Machine Learning Layer

5. Sentiment Analysis Layer
6. Visualization Layer

Figure 3.1: The architecture flow is as follows:



The architecture ensures smooth data flow from raw review collection to analytical visualization.

3.7 Data Flow Diagram (DFD)

Data Flow Diagrams are used to visually represent the flow of information within the system.

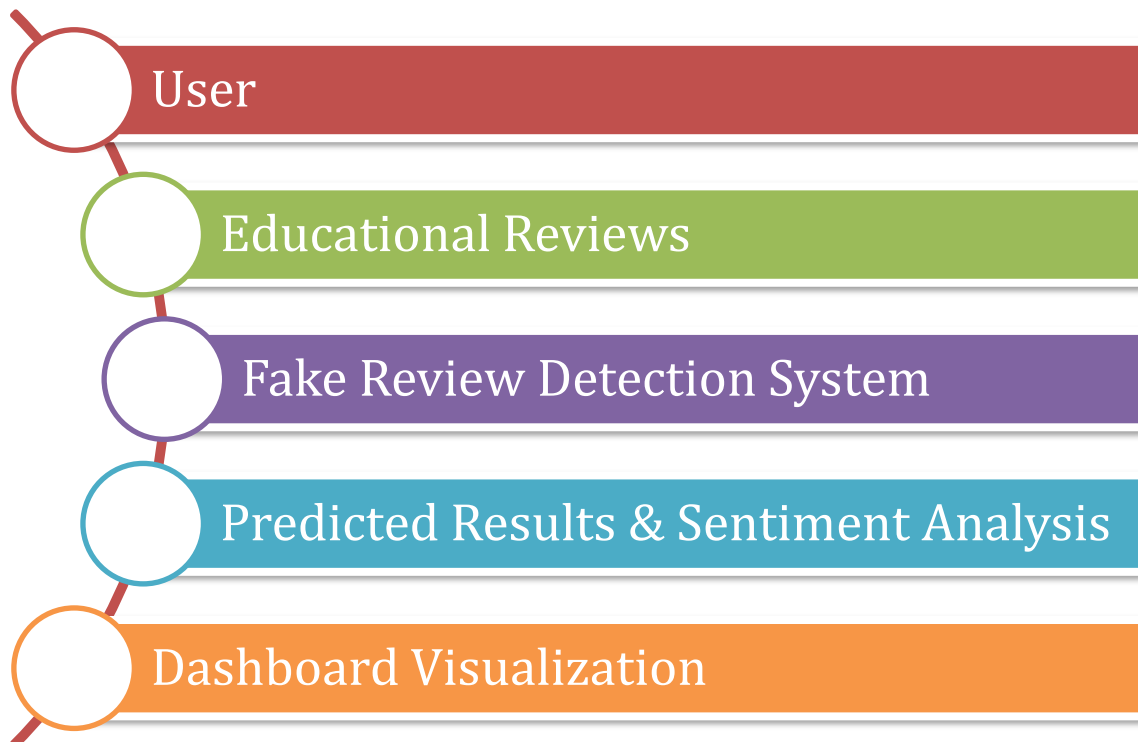
3.7.1 DFD Level 0

The Level 0 DFD provides a high-level overview of the system.

Main Components:

- User
- Review Dataset
- Fake Review Detection System
- Power BI Dashboard

Figure 3.2: Data Flow:



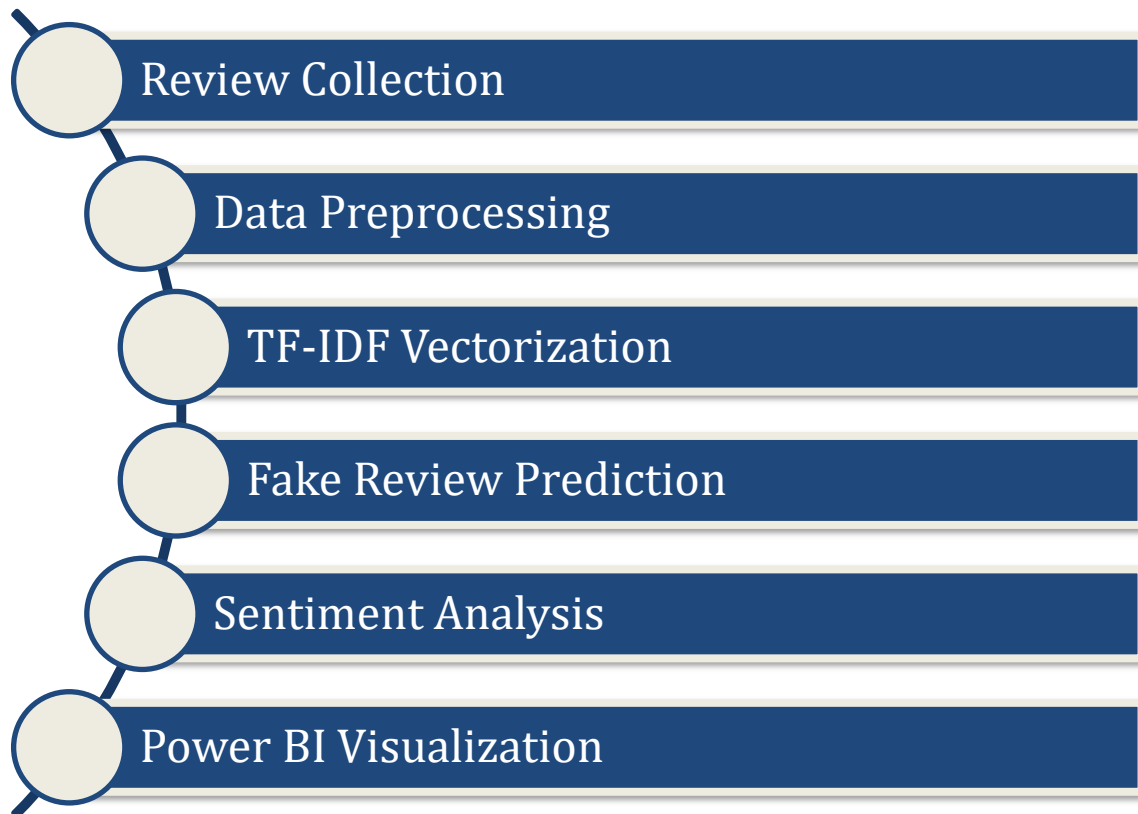
3.7.2 DFD Level 1

The Level 1 DFD explains internal processing stages in detail.

Modules Included:

1. Review Collection Module
2. Data Cleaning Module
3. Feature Extraction Module
4. Machine Learning Prediction Module
5. Sentiment Analysis Module
6. Dashboard Visualization Module

Figure 3.3: Process Flow:



3.8 Advantages of the Proposed System

The proposed system provides several advantages compared to traditional review analysis systems.

Automated Detection

The system automatically identifies potentially deceptive reviews using Machine Learning algorithms.

Educational Domain Focus

Unlike many existing systems, the proposed project specifically focuses on educational institutions.

Sentiment-Based Analysis

The integration of NLP techniques helps analyze emotional patterns within reviews.

Interactive Visualization

Power BI dashboards improve interpretability and allow users to explore review insights interactively.

Scalability

The system can be extended in the future to support:

- Larger datasets
 - Additional review platforms
 - Advanced AI models
-

3.9 Summary

This chapter presented the system analysis of the proposed fake review detection system. The analysis included functional requirements, non-functional requirements, user requirements, feasibility studies, system architecture, and data flow analysis.

The proposed system was found to be technically, economically, and operationally feasible. The architecture and workflow provide a structured approach for detecting fake reviews and generating analytical insights for educational review platforms.

Chapter 4: System Design

4.1 Introduction

System design is one of the most important stages in software development because it transforms system requirements into a structured technical solution. The system design phase focuses on defining architecture, workflows, data structures, modules, interfaces, and interactions among system components.

In this project, the system design focuses on creating an intelligent fake review detection framework using Machine Learning, Natural Language Processing, and Power BI visualization techniques. The design ensures efficient flow of data from review collection to prediction and dashboard visualization.

The proposed system is divided into multiple modules including:

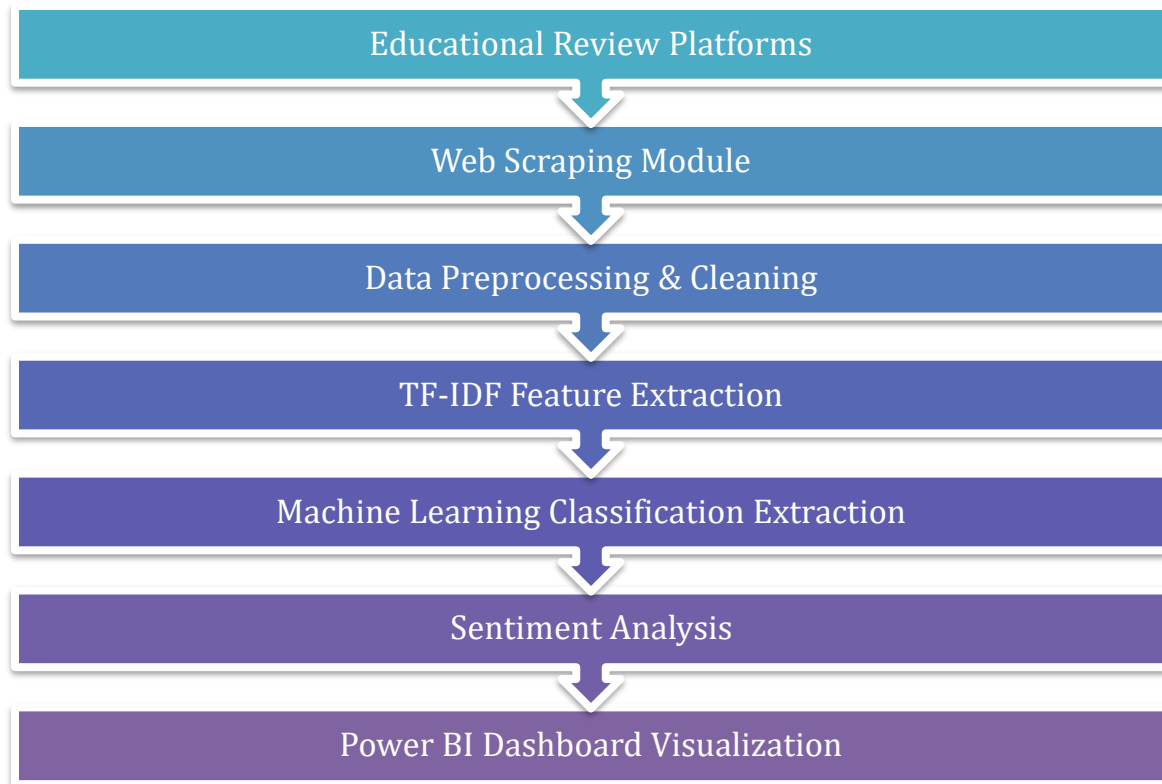
- Data Collection Module
- Data Preprocessing Module
- Feature Extraction Module
- Machine Learning Module
- Sentiment Analysis Module
- Dashboard Visualization Module

This chapter explains the UML diagrams, system architecture, database design, workflow models, and interface structures used in the project.

4.2 System Architecture Design

The proposed fake review detection system follows a layered architecture where each module performs a specific task in the overall workflow.

Figure 4.1: System Architecture Diagram



Explanation

The system begins with collecting educational reviews from publicly available platforms such as Collegedunia, Career360, and Google Reviews. The collected data is cleaned and preprocessed using NLP techniques.

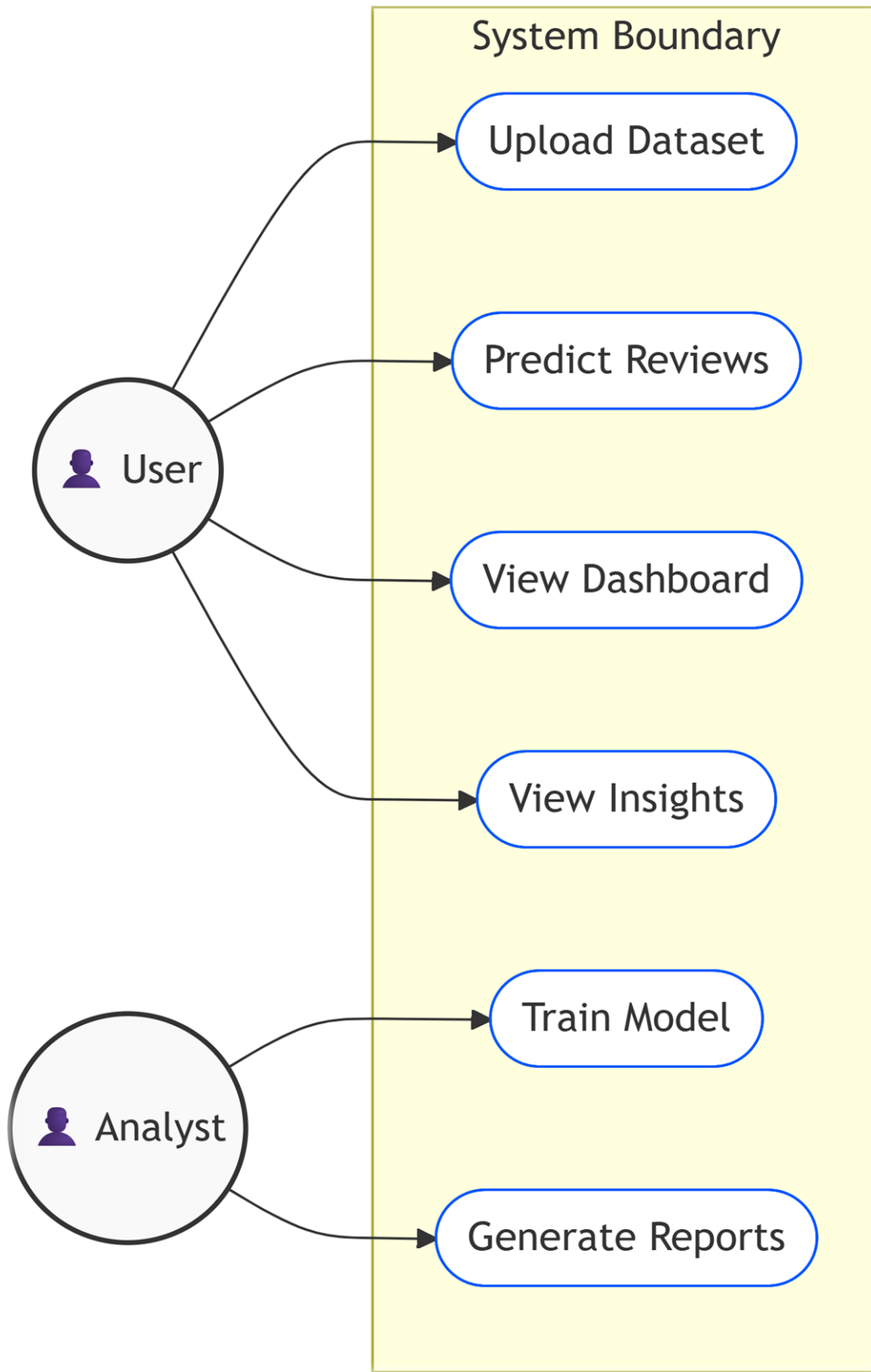
The cleaned text is converted into numerical vectors using TF-IDF vectorization. A Logistic Regression model is then used to classify reviews as fake or genuine. Sentiment analysis is performed using VADER sentiment analysis, and the final outputs are visualized through Power BI dashboards.

4.3 UML Diagrams

Unified Modeling Language (UML) diagrams help visualize system functionality, user interactions, workflows, and data relationships.

4.3.1 Use Case Diagram

Figure 4.2: Use Case Diagram



Actors:

- User
- Admin/Analyst

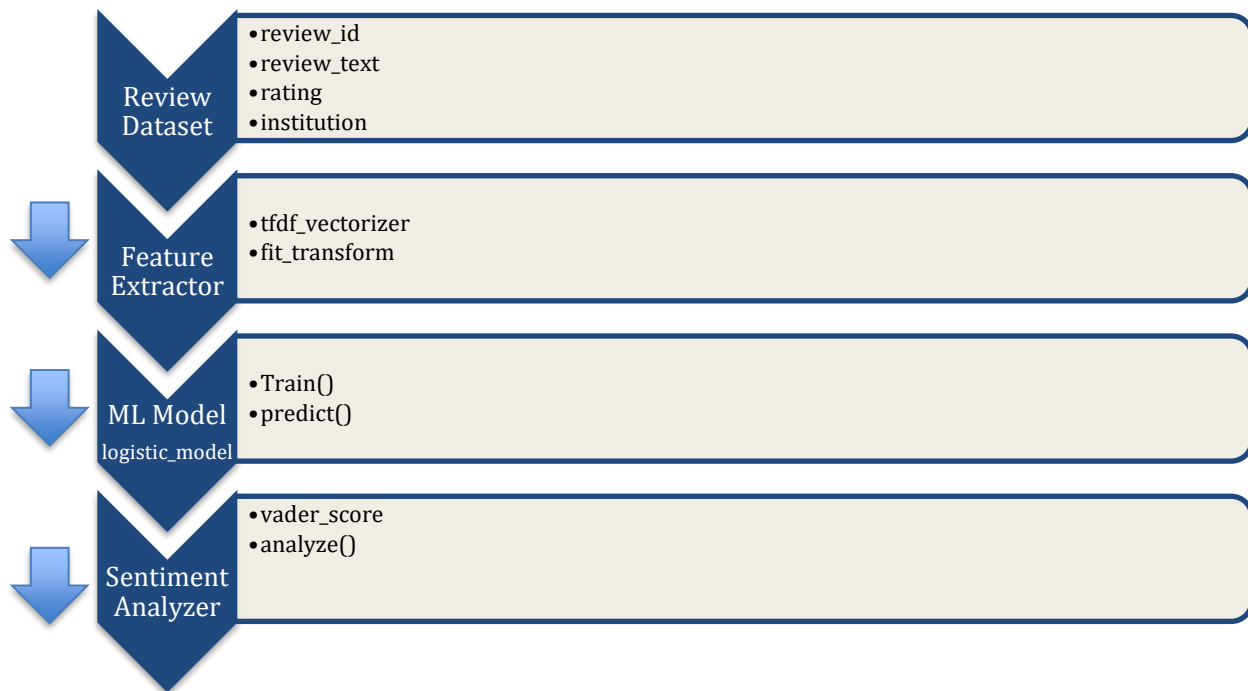
Use Cases:

- Upload Dataset
- Scrape Reviews
- Clean Data
- Train Model
- Predict Fake Reviews
- Perform Sentiment Analysis
- Generate Dashboard
- View Insights

Explanation

The Use Case Diagram represents interactions between users and the fake review detection system. Users can upload or analyze datasets, while the system performs preprocessing, machine learning classification, sentiment analysis, and dashboard generation.

4.3.2 Class Diagram**Figure 4.3: Class Diagram**



Classes Included:

1. Review Dataset
2. Preprocessing Module
3. Feature Extraction Module
4. ML Model
5. Sentiment Analyzer
6. Dashboard Module

Explanation

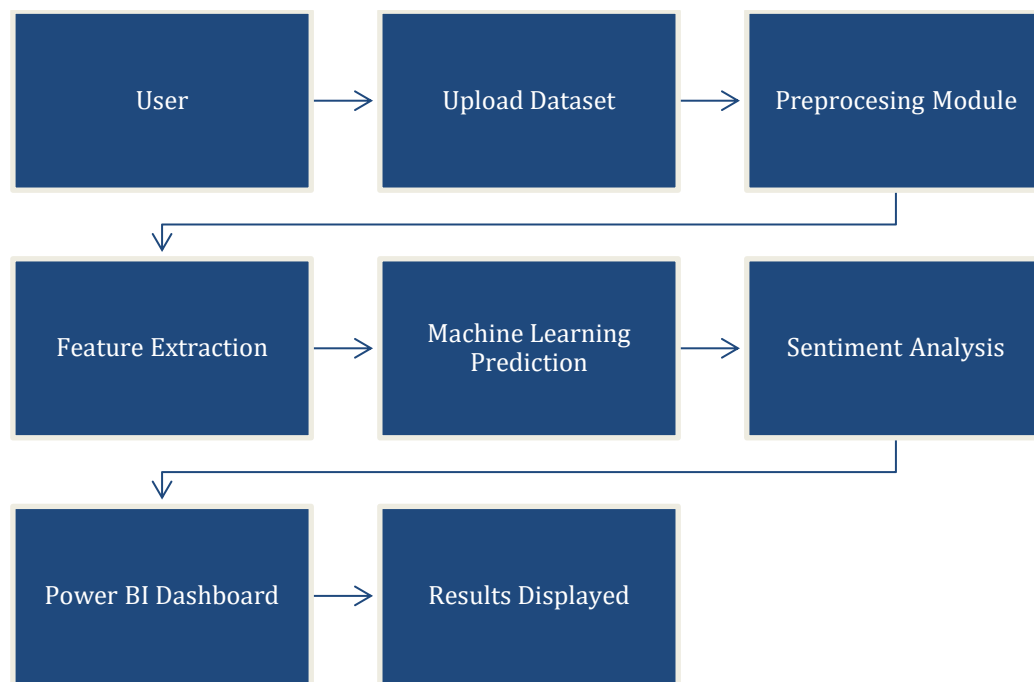
The Class Diagram illustrates the structural relationships between system modules.

- ReviewDataset stores collected review information.
- PreprocessingModule handles cleaning operations.
- FeatureExtractionModule converts text into TF-IDF vectors.
- MLModel predicts fake reviews.
- SentimentAnalyzer calculates sentiment scores.
- DashboardModule visualizes analytical results.

4.3.3 Sequence Diagram

Figure 4.4: Sequence Diagram

Sequence Flow



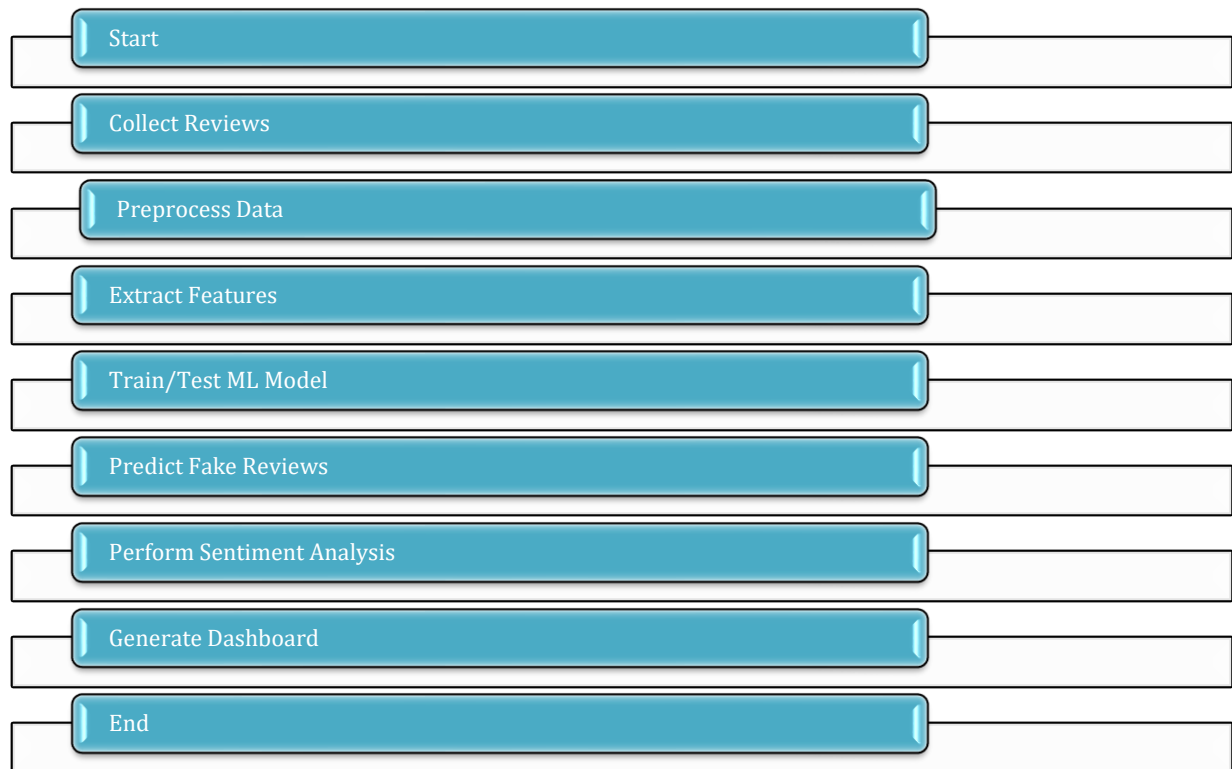
Explanation

The Sequence Diagram shows the order of interactions between system components during fake review analysis. The process starts when the user uploads review data and ends with dashboard visualization of insights.

4.3.4 Activity Diagram

Figure 4.5: Activity Diagram

Workflow



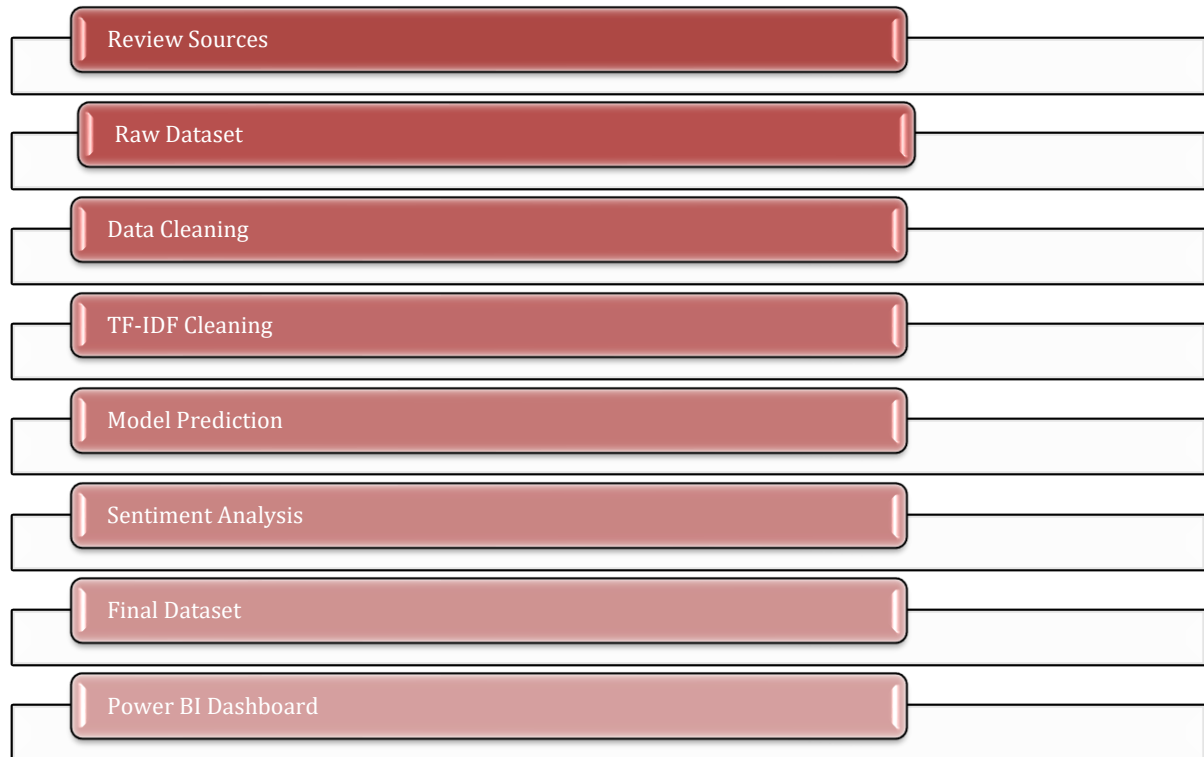
Explanation

The Activity Diagram represents the workflow of the fake review detection system from data collection to visualization.

4.4 Data Flow Design

The data flow design explains how data moves through different modules of the system.

Figure 4.6: Data Flow Workflow

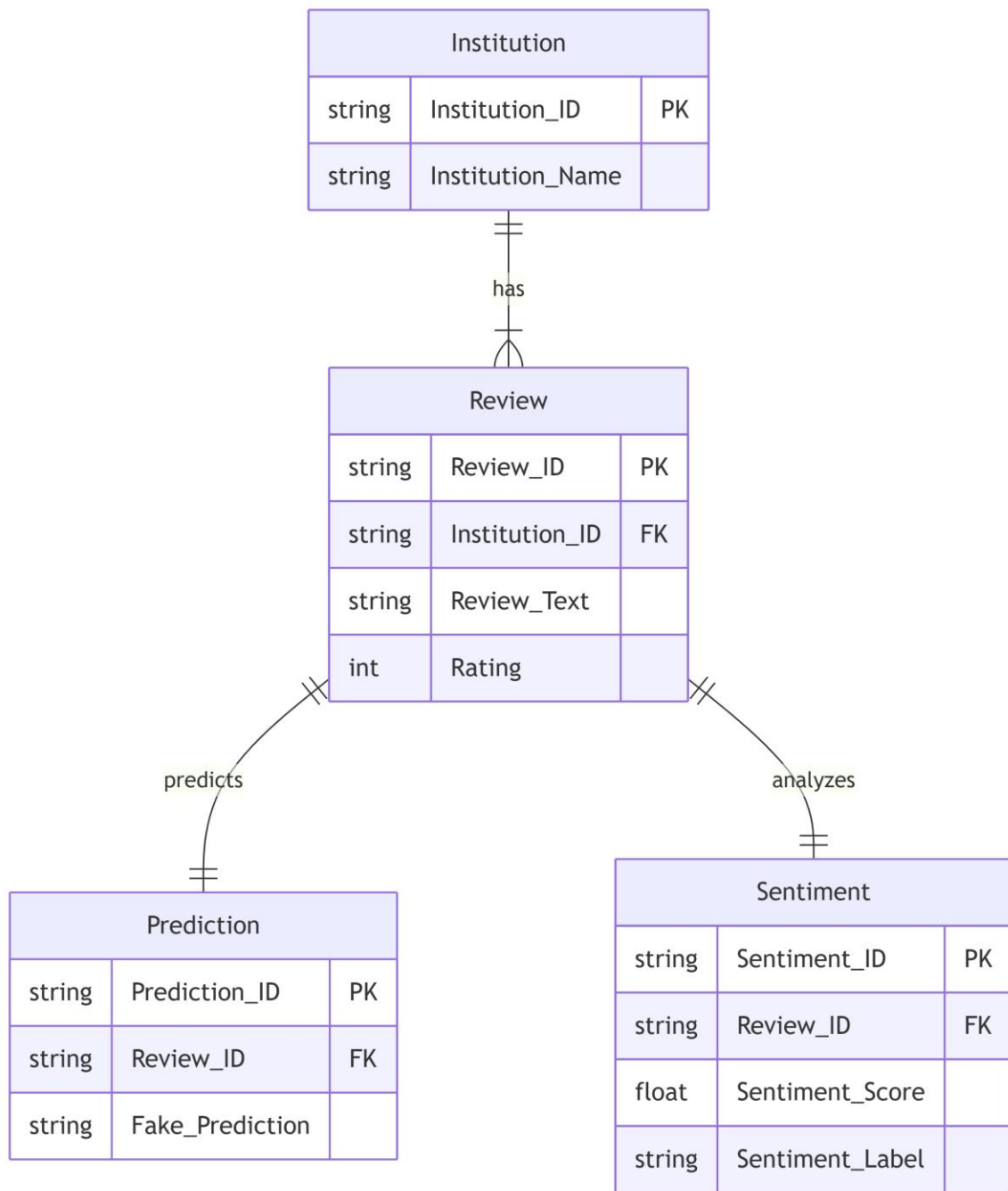


Explanation

The system processes raw educational review data and transforms it into structured analytical outputs using Machine Learning and NLP techniques.

4.5 ER Diagram (Entity Relationship Diagram)

Figure 4.7: ER Diagram



Entities Included

1. Institution
2. Review
3. Prediction
4. Sentiment

Relationships

- One Institution can have multiple Reviews.
- Each Review has one Fake Prediction result.
- Each Review has one Sentiment category.

Explanation

The ER Diagram defines relationships between entities within the review dataset.

4.6 Database Schema Design

The database schema represents the structure of the dataset used in the project.

Main Dataset Table: education_reviews_analysis

Field Name	Data Type	Description
Institution_Name	Text	Name of institution
Ratings	Float	Review rating
Reviews	Text	Review content
Clean_Review	Text	Preprocessed review
Fake_Prediction	Text	Fake/Genuine label
Sentiment_Score	Float	Sentiment polarity score
Sentiment	Text	Positive/Neutral/Negative

Explanation

The dataset stores both raw and processed review information required for analysis and visualization.

4.7 Table Structures

Table 4.1: Institution Review Table

Column Name	Type	Purpose
Institution_Name	VARCHAR	Stores institution name
Ratings	FLOAT	Stores review rating
Reviews	TEXT	Stores raw review
Clean_Review	TEXT	Stores cleaned review
Fake_Prediction	VARCHAR	Prediction output
Sentiment	VARCHAR	Sentiment category

4.8 API Design (If Applicable)

This project primarily functions as an analytical Machine Learning system and does not currently include external API integration.

However, future versions of the system may integrate APIs for:

- Real-time review scraping
- Automated prediction services
- Dashboard updates
- Cloud deployment

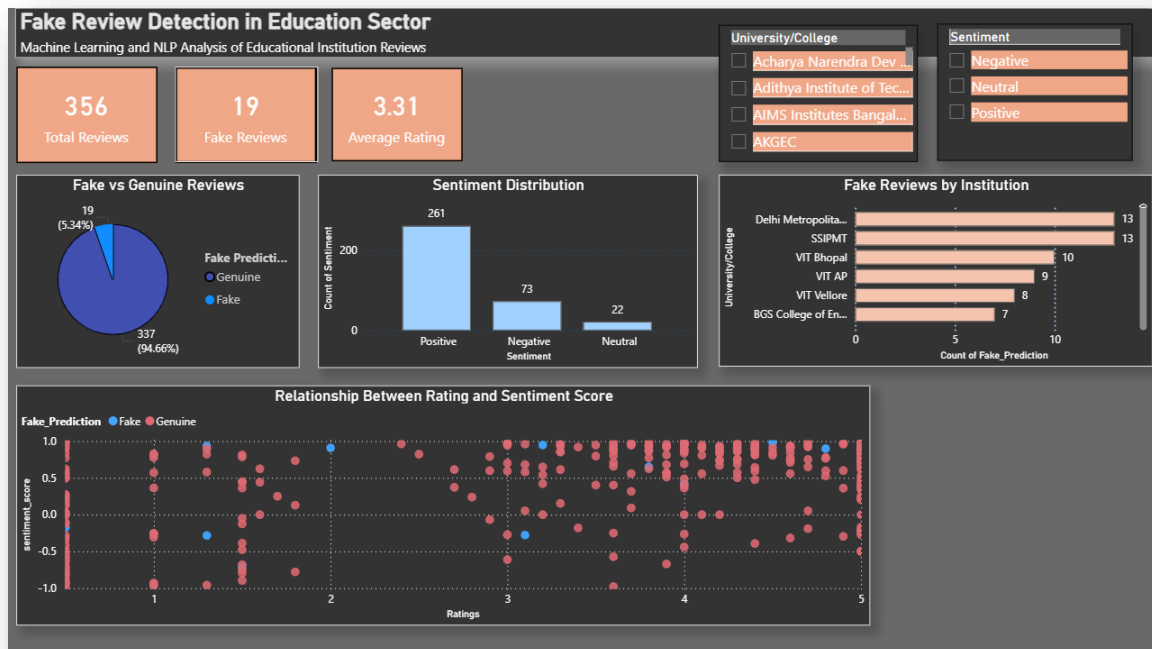
Possible API endpoints may include:

Endpoint	Purpose
/predict	Predict fake review
/sentiment	Perform sentiment analysis
/dashboard	Fetch dashboard insights

4.9 UI/UX Wireframe Design

The user interface was designed to provide simple and interactive visualization of review analysis.

Figure 4.8: Dashboard Wireframe



Dashboard Components

- KPI Cards
 - Total Reviews
 - Fake Reviews
 - Average Rating
- Pie Chart
 - Fake vs Genuine Reviews
- Sentiment Distribution Chart
- Fake Reviews by Institution
- Scatter Plot
 - Ratings vs Sentiment Score
- Interactive Filters

- Institution filter
- Sentiment filter

Explanation

The dashboard interface enables users to interactively explore fake review patterns, sentiment distributions, and institutional insights.

4.10 Design Considerations

Several important design considerations were followed during system development.

Simplicity

The dashboard and workflow were designed to be understandable for both technical and non-technical users.

Scalability

The architecture allows future integration of:

- Additional review platforms
 - Larger datasets
 - Advanced Machine Learning models
-

Modularity

The system modules are independent, making future updates and maintenance easier.

Visualization Focus

Special attention was given to creating meaningful visual analytics using Power BI dashboards.

4.11 Advantages of the Designed System

The proposed system design provides multiple advantages.

- Automated fake review detection
 - Sentiment-driven analysis
 - Education-specific analytical approach
 - Interactive dashboard visualization
 - Modular architecture for future scalability
-

4.12 Summary

This chapter presented the system design of the proposed fake review detection system. The chapter covered UML diagrams, workflow architecture, database schema, data flow structure, wireframe design, and analytical modules.

The proposed design ensures efficient integration of Machine Learning, NLP, and Power BI components for detecting deceptive educational reviews and generating strategic insights.

Chapter 5: System Implementation

5.1 Introduction

The development of the proposed fake review detection system was carried out using Python due to its extensive ecosystem for data science, machine learning, and natural language processing. Python provides a wide range of open-source libraries that simplify tasks such as data preprocessing, feature extraction, model training, evaluation, and visualization. The implementation was performed using Jupyter Notebook within the Anaconda distribution, providing an interactive environment suitable for experimentation and iterative development.

The machine learning model was developed using the Scikit-learn library, while Natural Language Processing tasks were implemented using the Natural Language Toolkit (NLTK). Data manipulation and preprocessing were performed using the Pandas library, whereas Power BI Desktop was used for creating interactive dashboards to visualize the final results.

The entire implementation process consisted of several stages including dataset collection, data preprocessing, education-specific filtering, feature extraction, machine learning model training, fake review prediction, sentiment analysis, and dashboard development. Each module was tested independently before integrating them into the final workflow.

5.2 Development Environment

The project utilized multiple software tools and technologies to build a complete fake review detection pipeline. Python served as the primary programming language because of its rich ecosystem for machine learning and data analysis. Jupyter Notebook was selected as the integrated development environment due to its support for interactive execution and visualization.

Scikit-learn was employed to implement the Logistic Regression classifier and TF-IDF vectorization. The NLTK library provided text preprocessing functions and the VADER sentiment analyzer for sentiment classification. Pandas was used extensively for reading datasets, cleaning textual information, transforming data structures, and exporting processed datasets for visualization.

Power BI Desktop was selected for dashboard development because of its ability to create interactive reports and business intelligence visualizations. The processed dataset generated from Python was directly imported into Power BI, enabling dynamic exploration of fake review predictions, sentiment distribution, institution-wise analysis, and rating trends.

Table 5.1: Development Environment Used

Component	Description
Programming Language	Python
IDE / Environment	Jupyter Notebook
Visualization Tool	Power BI Desktop
Version Control	Git & GitHub
Operating System	Windows 10

Explanation

Jupyter Notebook was used because it provides an interactive environment suitable for Machine Learning experimentation and NLP analysis. Power BI Desktop was used for dashboard creation and visualization of analytical insights.

GitHub was used for maintaining project files, datasets, notebooks, and documentation.

5.3 Tools and Technologies Used

Several tools and libraries were used during implementation.

Table 5.2: Tools and Technologies Used

Technology	Purpose
Python	Core programming language
Pandas	Data preprocessing
NumPy	Numerical operations
Scikit-learn	Machine Learning

NLTK	NLP and sentiment analysis
TF-IDF	Feature extraction
Logistic Regression	Fake review classification
Power BI	Dashboard visualization
GitHub	Version control

5.4 Hardware Requirements

The project was implemented on a standard personal computer with the following specifications.

Hardware Component	Specification
Processor	Intel Core i5 or above
RAM	8 GB
Storage	256 GB SSD
Internet Connection	Required for scraping
Display	Standard Monitor

Explanation

The project does not require high-end hardware because the dataset size is moderate and the machine learning algorithms used are lightweight.

5.5 Software Requirements

Software	Purpose
Python 3.x	Programming
Jupyter Notebook	Development Environment
Power BI Desktop	Dashboard Development
Git	Version Control
VS Code (Optional)	Code Editing

5.6 Module-Wise Implementation

The proposed system was developed using a modular architecture. Each module performs a specific task and passes its output to the next stage, making the system easier to maintain and extend.

The implementation consists of the following modules:

- Dataset Collection Module
- Data Cleaning Module
- Education Keyword Filtering Module
- Feature Extraction Module
- Machine Learning Module
- Fake Review Prediction Module
- Sentiment Analysis Module
- Dashboard Generation Module

Each module is discussed in detail in the following sections.

5.6.1 Data Collection Module

The first implementation stage involved collecting educational review data from publicly available platforms.

Sources Used

- Collegedunia
- Career360
- Google Reviews

The collected dataset contained:

- Institution names
- Ratings
- Review text

- Source information

Additionally, fake review datasets were downloaded from:

- Kaggle
- Yelp
- HuggingFace

These datasets were used for supervised model training.

Implementation Logic

The educational reviews were scraped and stored in CSV format. Since educational fake review datasets were not publicly available, cross-domain fake review datasets were used for model training.

5.6.2 Data Preprocessing Module

Raw textual review data contains noise and inconsistencies that reduce model performance.

The preprocessing implementation included:

- Lowercase conversion
- Punctuation removal
- Stopword removal
- Duplicate removal
- Text normalization

Sample Code Snippet

```
def clean_text(text):  
    text = text.lower()  
    text = re.sub(r'^a-zA-Z\s|', '', text)  
    tokens = text.split()  
    tokens = [word for word in tokens if word not in stop_words]  
    return " ".join(tokens)
```

Explanation

The preprocessing function converts text into a normalized format suitable for machine learning analysis.

5.6.3 Domain Adaptation Implementation

One major implementation challenge was domain mismatch because the fake review datasets belonged to industries such as hotels and products.

To improve educational relevance, keyword filtering was implemented.

Sample Logic

```
education_keywords = [  
    "college",  
    "university",  
    "faculty",  
    "campus",  
    "students",  
    "placement"  
]
```

Explanation

Only reviews containing education-related terms were retained to improve alignment with the educational domain.

5.6.4 Feature Extraction Module

Machine Learning algorithms cannot process raw text directly. Therefore, textual reviews were converted into numerical vectors using TF-IDF vectorization.

Sample Code Snippet

```
vectorizer = TfidfVectorizer(max_features=5000)  
X = vectorizer.fit_transform(df["clean_review"])
```

Explanation

TF-IDF calculates the importance of words across the dataset and converts reviews into feature vectors suitable for classification.

5.6.5 Machine Learning Model Module

The fake review detection model was implemented using Logistic Regression.

Model Training Process

1. Split dataset into training and testing sets
2. Train Logistic Regression model
3. Predict fake reviews
4. Evaluate performance

Sample Code Snippet

```
model = LogisticRegression()  
model.fit(X_train, y_train)  
predictions = model.predict(X_test)
```

Explanation

Logistic Regression was selected because it performs efficiently for binary text classification problems and works effectively with TF-IDF features.

5.6.6 Sentiment Analysis Module

Sentiment analysis was implemented using the VADER sentiment analyzer from the NLTK library.

Sample Code Snippet

```
sia = SentimentIntensityAnalyzer()  
  
df["sentiment_score"] = df["Reviews"].apply(  
    lambda x: sia.polarity_scores(str(x))["compound"]  
)
```

Explanation

The VADER analyzer assigns sentiment scores to reviews and classifies them into:

- Positive
- Neutral
- Negative

This helps identify emotional patterns within review data.

5.6.7 Dashboard Visualization Module

The final processed dataset was imported into Power BI for dashboard creation.

The dashboard included:

- KPI cards
- Pie charts
- Bar charts
- Scatter plots
- Institution-level analysis
- Sentiment analysis

Dashboard Features

- Interactive filters
- Dynamic charts
- Real-time analytical visualization

Explanation

Power BI enabled visual interpretation of fake review patterns and sentiment trends.

5.7 Key Algorithms Used

Several algorithms and analytical techniques were used in the project.

Algorithm / Technique	Purpose
Logistic Regression	Fake review classification
TF-IDF	Text vectorization
VADER	Sentiment analysis
NLP Preprocessing	Text cleaning

5.8 Security Features

Although the project is analytical in nature, several security considerations were followed.

Data Integrity

Processed datasets were stored separately to avoid accidental modification of raw datasets.

Safe Data Handling

Only publicly available educational reviews were collected and analyzed.

Version Control Security

GitHub was used for maintaining project versions and preventing accidental code loss.

5.9 Version Control Using GitHub

GitHub was used to manage project files, datasets, notebooks, reports, and dashboards.

Benefits of GitHub usage include:

- Backup of project files
- Version tracking
- Collaboration support
- Easy project sharing

The GitHub repository contains:

- Python notebooks
- Datasets

- Dashboard screenshots
 - Reports and presentations
-

Jupyter Notebook Implementation Screenshot

5.10 Dataset Collection Module

The first phase of implementation involved collecting two different categories of datasets. Since no publicly available labeled fake review dataset exists specifically for educational institutions, two complementary datasets were prepared.

The first dataset consisted of labeled fake and genuine reviews obtained from the Hugging Face Fake Reviews Dataset and the Yelp Fake Review Dataset. These datasets originally contained reviews from domains such as hotels, restaurants, electronics, movies, and consumer products.

The second dataset consisted of genuine educational reviews collected through web scraping from platforms including Collegedunia, Career360, and Google Reviews. These reviews contained information related to placements, faculty, infrastructure, hostel facilities, academic curriculum, and student experiences.

This dual-dataset strategy enabled the development of a domain-adapted fake review detection model while addressing the scarcity of educational fake review datasets.

Figure 5.1: Dataset Loading

```
[2]:  
  
# =====  
# 2 LOAD FAKE REVIEW DATASET (TRAINING DATA)  
# =====  
  
df = pd.read_csv("fake reviews dataset.csv")  
  
print("Original dataset shape:", df.shape)  
  
review_col = "review"  
label_col = "label"  
  
Original dataset shape: (40432, 4)
```

Explanation:

Figure 5.1 illustrates the successful loading of the training dataset into the Python environment. The original dataset contained 40,432 reviews collected from multiple domains. The dataset was imported using the Pandas library, allowing efficient preprocessing and transformation in subsequent implementation stages.

5.11 Data Cleaning Module

Before training the machine learning model, extensive preprocessing was performed to improve the quality of textual data. Online reviews often contain hyperlinks, punctuation, numbers, emojis, inconsistent capitalization, and irrelevant symbols that can negatively affect model performance.

A custom preprocessing function was developed to clean the review text. The function converts all text to lowercase, removes URLs, numerical values, punctuation marks, and stop words using the NLTK stopwords corpus. These operations reduce textual noise while preserving meaningful information for machine learning.

The cleaned reviews were stored in a new column named **clean_review**, which served as the input for feature extraction.

Figure 5.2 Text Cleaning Output



```

jupyter Fake-Review-Model Last Checkpoint: 2 months ago
File Edit View Run Kernel Settings Help Trusted
JupyterLab Python (venv_3.10)

def is_education_related(text):
    text = str(text).lower()
    return any(word in text for word in education_keywords)

df_filtered = df[df["clean_review"].apply(is_education_related)]

print("After education filtering:", df_filtered.shape)

# =====
# 6 FIX LABELS (OR / CG -> Genuine / Fake)
# =====

print("Original labels:", df_filtered[label_col].unique())

df_filtered[label_col] = df_filtered[label_col].map({
    "OR": "Genuine",
    "CG": "Fake"
})

print("Mapped labels:")
print(df_filtered[label_col].value_counts())

Original dataset shape: (40432, 4)
After education filtering: (2072, 5)
Original labels: ['CG' 'OR']
Mapped labels:
label
Genuine    1476
Fake        596
Name: count, dtype: int64

```

Explanation

Figure 5.2 demonstrates the preprocessing stage where raw review text is transformed into a normalized format suitable for machine learning. Cleaning significantly reduces vocabulary size and improves the consistency of textual features extracted by the TF-IDF vectorizer.

5.12 Education Domain Filtering

One of the most significant challenges encountered during the implementation was the absence of an education-specific fake review dataset. The available benchmark datasets primarily contained reviews from unrelated domains such as hotels, restaurants, and consumer products.

To address this issue, a domain adaptation strategy was implemented using keyword-based filtering. Reviews containing education-related terms such as **college**, **university**, **faculty**, **professor**, **placement**, **semester**, **course**, **students**, **hostel**, **laboratory**, and **campus** were retained, while unrelated reviews were discarded.

After filtering, the dataset size reduced from **40,432 reviews** to **2,072 education-related reviews**, significantly improving domain relevance while maintaining sufficient training data.

Figure 5.3 Education Dataset Filtering

```
# =====
# 6 FIX LABELS (OR / CG → Genuine / Fake)
# =====

print("Original labels:", df_filtered[label_col].unique())

df_filtered[label_col] = df_filtered[label_col].map({
    "OR": "Genuine",
    "CG": "Fake"
})

print("Mapped labels:")
print(df_filtered[label_col].value_counts())
```

```
Original dataset shape: (40432, 4)
After education filtering: (2072, 5)
Original labels: ['CG' 'OR']
Mapped labels:
label
Genuine    1476
Fake        596
Name: count, dtype: int64
```

Explanation

The filtering process ensured that the machine learning model learned linguistic patterns specific to educational institutions rather than unrelated industries. This improved the applicability of the trained model to real educational review data collected during the project.

5.13 Feature Extraction using TF-IDF

Text data cannot be directly processed by machine learning algorithms. Therefore, the cleaned reviews were converted into numerical vectors using the TF-IDF (Term Frequency–Inverse Document Frequency) vectorization technique.

A vocabulary size of 5,000 features was selected to balance computational efficiency and predictive performance. TF-IDF assigns higher weights to informative words while reducing the influence of commonly occurring words.

The transformed feature matrix served as the input for the Logistic Regression classifier.

Figure 5.4 TF-IDF Feature Extraction

```
# =====
# 14 TF-IDF TRANSFORM
# =====

scrapped_features = vectorizer.transform(scraped["clean_review"])

# =====
# 15 PREDICT FAKE REVIEWS
# =====

scraped["Fake_Prediction"] = model.predict(scrapped_features)

print("Prediction distribution:")
print(scraped["Fake_Prediction"].value_counts())
```

Accuracy: 0.908433734939759

	precision	recall	f1-score	support
Fake	0.86	0.82	0.84	119
Genuine	0.93	0.95	0.94	296
accuracy			0.91	415
macro avg	0.89	0.88	0.89	415
weighted avg	0.91	0.91	0.91	415

Scraped dataset size: (356, 4)
Prediction distribution:
Fake_Prediction
Genuine 337
Fake 19
Name: count, dtype: int64

Explanation

The TF-IDF transformation converts textual reviews into sparse numerical vectors. These vectors capture the importance of individual words within each review, allowing the machine learning model to recognize linguistic patterns associated with deceptive reviews.

The technique assigns numerical weights to words based on their frequency within a document and across the complete dataset. Frequently occurring words that appear in almost every review receive lower importance, whereas unique and informative words receive higher weights. This representation enables the Logistic Regression classifier to distinguish between genuine and deceptive reviews by learning linguistic patterns associated with educational feedback. Compared with simple Bag-of-Words models, TF-IDF reduces the influence of common terms while emphasizing discriminative vocabulary, resulting in improved text representation and better classification performance.

5.14 Training and Testing Data Preparation

Before training the machine learning model, the filtered educational review dataset was divided into training and testing subsets. Splitting the dataset allows the model to learn from one portion of the data while evaluating its performance on previously unseen reviews. This approach helps measure the generalization capability of the classifier and minimizes the risk of overfitting.

The cleaned review text was first converted into numerical feature vectors using the TF-IDF vectorizer. The resulting feature matrix was then divided into training and testing datasets using the `train_test_split()` function available in the Scikit-learn library. In this project, 80% of the dataset was allocated for training, while the remaining 20% was reserved for testing. A fixed random state of 42 was used to ensure reproducibility of the experimental results.

The training dataset was used to build the Logistic Regression model, whereas the testing dataset was used exclusively for evaluating prediction accuracy. Maintaining separate datasets for training and evaluation ensures that the reported performance accurately reflects the model's ability to classify unseen reviews rather than memorizing the training data.

Figure 5.5: Train-Test Split

```
[4]: # =====  
# 7 PREPARE TRAINING DATA  
# =====  
  
X = df_filtered["clean_review"]  
y = df_filtered[label_col]  
  
[5]: # =====  
# 8 TF-IDF VECTORIZATION  
# =====  
  
vectorizer = TfidfVectorizer(max_features=5000)  
  
X_vectorized = vectorizer.fit_transform(X)  
  
[7]: # =====  
# 9 TRAIN TEST SPLIT  
# =====  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X_vectorized,  
    y,  
    test_size=0.2,  
    random_state=42  
)  
  
# =====  
# 10 TRAIN MACHINE LEARNING MODEL  
# =====
```

Explanation

Figure 5.5 illustrates the implementation of the train-test split process. Dividing the dataset before model training ensures that the classifier is evaluated objectively using independent data, providing a realistic estimate of model performance.

5.15 Machine Learning Model Training

The Logistic Regression algorithm was selected as the primary machine learning classifier because of its simplicity, computational efficiency, and proven effectiveness in binary text classification problems. Logistic Regression estimates the probability that a review belongs to either the "Fake" or "Genuine" category by learning relationships between textual features extracted through TF-IDF vectorization.

The model was initialized with a maximum iteration limit of 1000 to ensure convergence during optimization. Since the filtered dataset contained more genuine reviews than fake reviews, the `class_weight="balanced"` parameter was applied to compensate for class imbalance. This adjustment helps prevent the model from becoming biased toward the majority class.

During training, the algorithm learned linguistic patterns commonly associated with deceptive and authentic reviews. These patterns included differences in vocabulary usage, writing style, and contextual expressions observed within the training dataset.

Figure 5.6: Logistic Regression Model Training

```
# =====  
# 10 TRAIN MACHINE LEARNING MODEL  
# =====  
  
model = LogisticRegression(max_iter=1000, class_weight="balanced")  
model.fit(X_train, y_train)  
  
# =====  
# 11 EVALUATE MODEL  
# =====  
  
pred = model.predict(X_test)  
  
print("Accuracy:", accuracy_score(y_test, pred))  
print(classification_report(y_test, pred))  
  
# =====  
# 12 LOAD SCRAPED EDUCATION REVIEWS  
# =====  
  
scraped = pd.read_csv("institutes.csv", encoding="latin-1")  
print("Scraped dataset size:", scraped.shape)  
  
# =====  
# 13 CLEAN SCRAPED REVIEWS  
# =====
```

Explanation

Figure 5.6 demonstrates the successful training of the Logistic Regression classifier. The model learns from TF-IDF feature vectors and generates a decision boundary capable of distinguishing between fake and genuine educational reviews.

5.16 Model Performance Evaluation

After training the classifier, the model was evaluated using the testing dataset. Multiple evaluation metrics were considered to provide a comprehensive assessment of classification performance. These included Accuracy, Precision, Recall, and F1-score.

The proposed model achieved an overall accuracy of **90.84%**, indicating that it correctly classified the majority of unseen reviews. For the fake review class, the model achieved a

precision of **86%**, a recall of **82%**, and an F1-score of **84%**. The genuine review class achieved even higher performance with a precision of **93%**, recall of **95%**, and F1-score of **94%**.

These results demonstrate that the proposed machine learning model performs effectively on educational review data despite being trained using a domain-adapted dataset.

Table 5.2 Model Evaluation Metrics

Metric	Fake Reviews	Genuine Reviews
Precision	86%	93%
Recall	82%	95%
F1-Score	84%	94%
Overall Accuracy	90.84%	

The Logistic Regression classifier achieved an accuracy of 90.84%, indicating that the proposed model correctly classified the majority of educational reviews. The high accuracy demonstrates that TF-IDF feature extraction combined with Logistic Regression effectively captures linguistic characteristics associated with deceptive reviews despite the limited availability of education-specific training data.

Figure 5.7: Classification Report

```
# 15 PREDICT FAKE REVIEWS
# =====
scraped["Fake_Prediction"] = model.predict(scraped_features)
print("Prediction distribution:")
print(scraped["Fake_Prediction"].value_counts())
```

Accuracy:	0.908433734939759			
	precision	recall	f1-score	support
Fake	0.86	0.82	0.84	119
Genuine	0.93	0.95	0.94	296
accuracy			0.91	415
macro avg	0.89	0.88	0.89	415
weighted avg	0.91	0.91	0.91	415

Scraped dataset size: (356, 4)
Prediction distribution:
Fake_Prediction
Genuine 337
Fake 19
Name: count, dtype: int64

Explanation

Figure 5.7 presents the evaluation results generated after testing the classifier. The high accuracy and balanced precision-recall values indicate that the proposed model effectively identifies deceptive educational reviews while minimizing false predictions.

5.17 Fake Review Prediction on Educational Dataset

Following successful model evaluation, the trained classifier was applied to the scraped educational review dataset. The reviews were collected from Collegedunia, Career360, and Google Reviews and represented authentic student experiences regarding academics, placements, faculty, campus facilities, and infrastructure.

Each review underwent the same preprocessing pipeline used during training, including text cleaning, stopword removal, and TF-IDF transformation. The trained Logistic Regression model then predicted whether each review was genuine or fake.

Out of **356 educational reviews**, **337 reviews were classified as genuine**, while **19 reviews were identified as potentially fake**. This indicates that approximately **5.34%** of the analyzed reviews exhibited linguistic characteristics similar to deceptive reviews found within the training dataset.

Figure 5.8: Fake Review Prediction

```
Scraped dataset size: (356, 4)
Prediction distribution:
Fake_Prediction
Genuine      337
Fake         19
Name: count, dtype: int64
```

Explanation

Figure 5.8 illustrates the prediction results generated by the trained classifier. The model successfully categorized each educational review into fake or genuine classes, providing the foundation for subsequent dashboard analysis.

5.18 Sentiment Analysis Implementation

In addition to fake review detection, sentiment analysis was incorporated to understand the emotional characteristics of student reviews. The VADER (Valence Aware Dictionary and Sentiment Reasoner) sentiment analyzer available in the NLTK library was selected because of its ability to evaluate textual sentiment efficiently.

Each review was assigned a compound sentiment score ranging from -1 to +1. Positive scores represented favorable opinions, negative scores represented dissatisfaction, and values close to zero indicated neutral sentiment.

The compound score was converted into three sentiment categories:

- Positive
- Neutral
- Negative

These categories were later visualized in Power BI to analyze the emotional distribution of educational reviews.

Figure 5.9: Sentiment Analysis Output

```
# =====
# 20 PREVIEW FINAL DATA
# =====

print(scraped.head())

Final dataset saved successfully

   University/College  Ratings \
0                GECA         3.6
1             GIMS Noida       4.1
2  Motilal Nehru Evening College  3.5
3  Narayana Engineering College Nellore  4.8
4  SRK Institute of Technology Vijayawada  5.0

   Reviews  Source \
0  there were 10 -15 faculty are in civil departm...  Collegedunia
1  The course curriculum is updated every year as...  Collegedunia
2  In regard to the relevancy of the course Im qu...  Collegedunia
3  Students are eligible for the placements from ...  Collegedunia
4  The faculty-to-student ratio is something like...  Collegedunia

   clean_review  Fake_Prediction \
0  faculty civil department faculty permanent tem...  Genuine
1  course curriculum updated every year per indus...  Genuine
2  regard relevancy course im quite fond subjects...  Genuine
3  students eligible placements semesters special...  Genuine
4  facultytostudent ratio something like faculty ...  Genuine

   sentiment_score  Sentiment
0          0.8621  Positive
1          0.5574  Positive
2          0.9465  Positive
3          0.5267  Positive
4          0.9682  Positive
```

Explanation

Figure 5.9 demonstrates the sentiment analysis process. The generated sentiment labels provide additional context for interpreting student opinions beyond fake review detection.

5.19 Final Dataset Generation and Power BI Integration

The final stage of implementation involved combining all generated information into a single structured dataset. The exported CSV file contained institution names, ratings, original review text, cleaned review text, fake review predictions, sentiment scores, and sentiment labels.

This processed dataset was imported into Microsoft Power BI Desktop, where multiple interactive dashboards were created. Visualizations included fake review distribution, sentiment analysis, institution-wise comparison, rating distribution, and key performance indicators. The dashboard enables users to explore review trends dynamically through interactive filters and charts.

Figure 5.10: Power BI Dashboard



Explanation

The Power BI dashboard transforms the processed review data into actionable insights, allowing educational institutions and prospective students to analyze review authenticity and sentiment efficiently.

5.20 Security and Ethical Considerations

The implementation was developed with attention to ethical data handling practices. Only publicly available reviews were collected from educational review platforms, and no personally identifiable information was stored or processed. The objective of the project was to identify potentially deceptive review patterns rather than make definitive judgments about individual reviewers.

The collected datasets were used solely for academic and research purposes. Data preprocessing focused on textual analysis without modifying or altering the original review content. Furthermore, the developed machine learning model serves as a decision-

support tool and should be used alongside human judgment when evaluating review authenticity.

5.11 Challenges Faced During Implementation

Several challenges were encountered during implementation.

Lack of Educational Fake Review Dataset

No dedicated fake review dataset for educational institutions was publicly available.

Solution

Cross-domain datasets were combined with scraped educational reviews.

Encoding Errors During Data Loading

Unicode decoding errors occurred while loading scraped datasets.

Solution

Alternative encoding methods such as:

```
encoding='latin-1'
```

were used to resolve the issue.

Domain Mismatch

Hotel and product reviews differed significantly from educational reviews.

Solution

Domain adaptation and keyword filtering techniques were implemented.

5.12 Summary

This chapter discussed the implementation of the fake review detection system using Machine Learning, NLP techniques, and Power BI visualization tools.

The implementation phase included dataset collection, preprocessing, TF-IDF vectorization, Logistic Regression classification, sentiment analysis using VADER, and dashboard visualization. The system successfully identified potentially deceptive educational reviews and generated meaningful analytical insights.

Chapter 6: Testing

6.1 Introduction

Testing is an essential phase in software development that ensures the system functions correctly, efficiently, and reliably according to the specified requirements. The purpose of testing is to identify errors, validate system performance, and ensure accurate execution of all modules.

In this project, testing was performed to verify the correctness of the fake review detection system, sentiment analysis workflow, machine learning predictions, and dashboard visualization. Multiple testing strategies were applied to validate data preprocessing, classification accuracy, dashboard functionality, and integration between system modules.

The testing phase ensured that the system was capable of processing educational reviews, detecting deceptive patterns, and generating meaningful analytical insights without major errors.

6.2 Testing Strategy

A structured testing strategy was followed to ensure proper validation of all components in the system.

The testing process included:

1. Unit Testing
2. Integration Testing
3. System Testing
4. Functional Testing
5. Output Validation

The testing strategy focused on verifying:

- Data preprocessing accuracy
- Machine Learning prediction correctness
- Sentiment analysis outputs
- Dashboard visualization functionality

- Dataset handling and export operations

The testing process was performed iteratively during development to identify and resolve implementation issues.

6.3 Unit Testing

Unit testing focuses on testing individual modules independently to ensure they function correctly.

Each major module of the project was tested separately.

6.3.1 Testing of Data Cleaning Module

The preprocessing module was tested to verify:

- Lowercase conversion
- Punctuation removal
- Stopword elimination
- Duplicate removal

Example

Input

"The College campus is GREAT!!!"

Expected Output

"college campus great"

Result

The preprocessing module successfully cleaned and normalized the review text.

6.3.2 Testing of TF-IDF Vectorization

The TF-IDF module was tested to verify successful conversion of textual data into numerical vectors.

Validation Performed

- Vocabulary generation
- Sparse matrix creation
- Feature extraction consistency

The vectorization process successfully transformed review text into machine-readable numerical representations.

6.3.3 Testing of Logistic Regression Model

The Machine Learning model was tested to ensure correct classification of reviews.

Validation Checks

- Model training
- Prediction generation
- Classification consistency

The model successfully generated fake and genuine review predictions.

6.3.4 Testing of Sentiment Analysis Module

The VADER sentiment analyzer was tested using sample positive and negative reviews.

Example Input

"The faculty and placements are excellent."

Expected Sentiment

Positive

Result

The module correctly identified positive sentiment based on sentiment scores.

6.4 Integration Testing

Integration testing verifies whether different modules work together correctly after integration.

In this project, integration testing was performed between:

- Data preprocessing and feature extraction
 - Feature extraction and machine learning model
 - Machine learning output and sentiment analysis
 - Processed dataset and Power BI dashboard
-

6.4.1 Preprocessing + Model Integration

The cleaned review dataset was passed into the TF-IDF vectorizer and Logistic Regression model.

Result

The integration successfully generated fake review predictions without data compatibility issues.

6.4.2 Model + Dashboard Integration

The prediction results were exported to CSV format and imported into Power BI.

Result

The Power BI dashboard correctly visualized fake review distributions and sentiment analysis results.

6.5 System Testing

System testing validates the complete functionality of the entire application workflow.

The complete system was tested from:

1. Review collection
2. Data preprocessing
3. Machine learning prediction
4. Sentiment analysis
5. Dashboard visualization

The system successfully processed educational reviews and generated analytical insights.

6.6 Functional Testing

Functional testing was performed to ensure all required functionalities worked correctly.

Functionality	Result
Review preprocessing	Passed
TF-IDF vectorization	Passed
Fake review prediction	Passed
Sentiment analysis	Passed
Dashboard visualization	Passed
Data export	Passed

6.7 Test Cases Table

Table 6.1: Test Cases

Test Case ID	Input	Expected Output	Actual Output	Status
TC-01	Raw review text	Cleaned review text	Cleaned successfully	Passed
TC-02	Review dataset	TF-IDF vectors	Vectors generated	Passed
TC-03	Review text	Fake/Genuine prediction	Prediction generated	Passed
TC-04	Positive review	Positive sentiment	Positive detected	Passed
TC-05	Dataset import to Power BI	Dashboard visualization	Dashboard displayed	Passed
TC-06	CSV export	Processed CSV file	Export successful	Passed

6.8 Bug Reports and Issues Faced

Several implementation issues were encountered during development and testing.

6.8.1 Unicode Decode Error

Issue

While loading scraped datasets using Pandas, encoding errors occurred.

Error Example

```
UnicodeDecodeError: 'utf-8' codec can't decode byte
```

Solution

Alternative encodings such as:

```
encoding='latin-1'
```

were used to resolve the issue.

6.8.2 Ratings Conversion Error

Issue

Ratings stored in formats such as:

```
4.5/5
```

could not be directly converted into float values.

Solution

String cleaning operations were applied before conversion.

Example

```
df["Ratings"] = df["Ratings"].str.replace("/5", "")
```

6.8.3 Domain Mismatch Problem

Issue

Fake review datasets from hotels and products did not align with educational reviews.

Solution

Keyword-based filtering and domain adaptation strategies were implemented.

6.8.4 Prediction Imbalance

Issue

Initially, the model predicted most reviews as genuine.

Solution

Additional fake review examples and balanced filtering techniques were used to improve predictions.

6.9 Performance Evaluation

The performance of the system was evaluated using Machine Learning classification outputs and analytical visualization quality.

Evaluation Metrics Considered

- Classification consistency
- Prediction reliability
- Sentiment classification quality
- Dashboard interpretability

The system successfully identified approximately 5.3% of reviews as potentially fake while maintaining meaningful sentiment classification.

6.10 Security and Reliability Testing

The project also considered basic security and reliability aspects.

Reliability Measures

- Backup of processed datasets
 - Version control using GitHub
 - Separate storage for raw and processed datasets
-

Data Safety

Only publicly available educational reviews were analyzed during testing and implementation.

6.11 Advantages of Testing

The testing process provided several benefits:

- Improved prediction reliability
- Reduced implementation errors
- Verified dashboard functionality
- Improved preprocessing quality
- Better integration between modules

6.12 Summary

This chapter discussed the testing methodologies applied during the development of the fake review detection system.

Different testing techniques such as unit testing, integration testing, and system testing were performed to validate preprocessing, Machine Learning predictions, sentiment analysis, and dashboard visualization.

The testing phase ensured that the system functioned correctly and generated meaningful analytical insights for identifying deceptive educational reviews.

Chapter 7: Results & Discussion

7.1 Introduction

The proposed system was evaluated using the educational review dataset collected from multiple online review platforms. After preprocessing and applying the trained Logistic Regression model, each review was classified as either Genuine or Fake. In addition, sentiment analysis was performed using the VADER sentiment analyzer to understand the emotional characteristics of the collected reviews. The processed data was finally visualized using Microsoft Power BI to provide interactive insights into fake review distribution, institutional ratings, and sentiment trends.

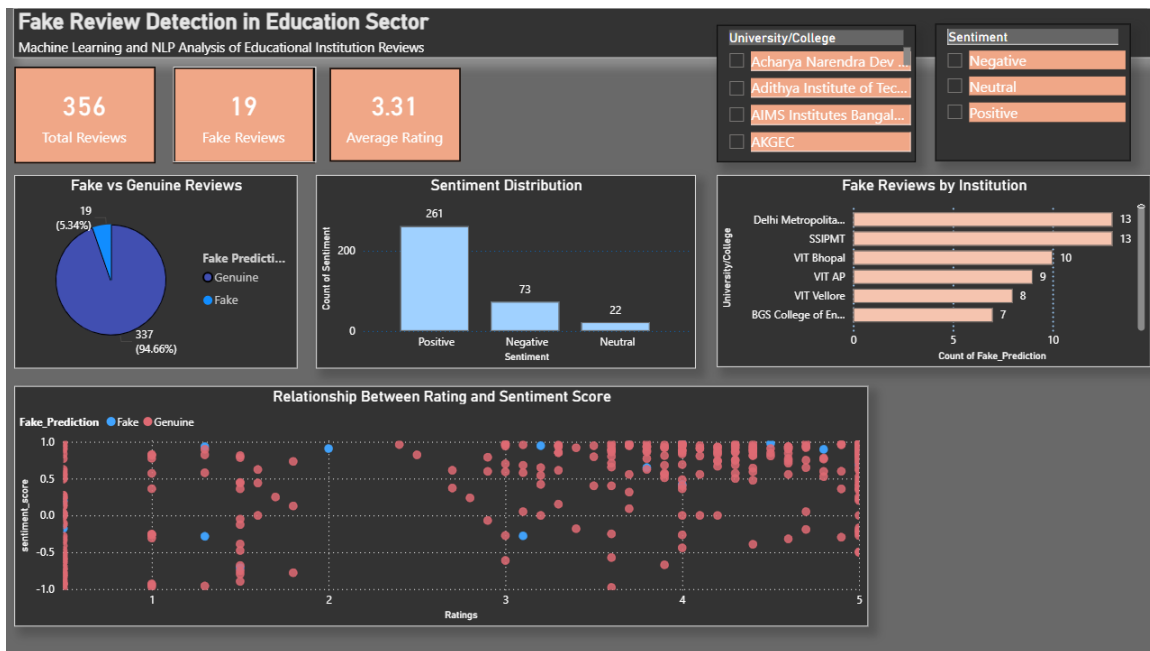
The experimental results demonstrate that the proposed approach successfully combines Machine Learning, Natural Language Processing, and Business Intelligence to identify potentially deceptive educational reviews while providing meaningful analytical insights.

7.2 Output Screenshots

The outputs generated during implementation and analysis are presented below.

7.2.1 Power BI Dashboard Output

Figure 7.1: Power BI Dashboard



Explanation

The Power BI dashboard displays key insights related to fake review detection and sentiment analysis.

The dashboard includes:

- Total Reviews
- Fake Reviews Detected
- Average Ratings
- Fake vs Genuine Review Distribution
- Sentiment Analysis
- Institution-Level Fake Review Analysis
- Rating vs Sentiment Scatter Plot

The interactive filters allow users to analyze reviews based on institutions and sentiment categories.

7.2.2 Fake Review Prediction Output

Figure 7.2: Machine Learning Prediction Output

```
[8]: # =====
# 11 EVALUATE MODEL
# =====

pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, pred))
print(classification_report(y_test, pred))
```

Accuracy: 0.908433734939759

	precision	recall	f1-score	support
Fake	0.86	0.82	0.84	119
Genuine	0.93	0.95	0.94	296
accuracy			0.91	415
macro avg	0.89	0.88	0.89	415
weighted avg	0.91	0.91	0.91	415

Explanation

The Machine Learning model classified reviews into:

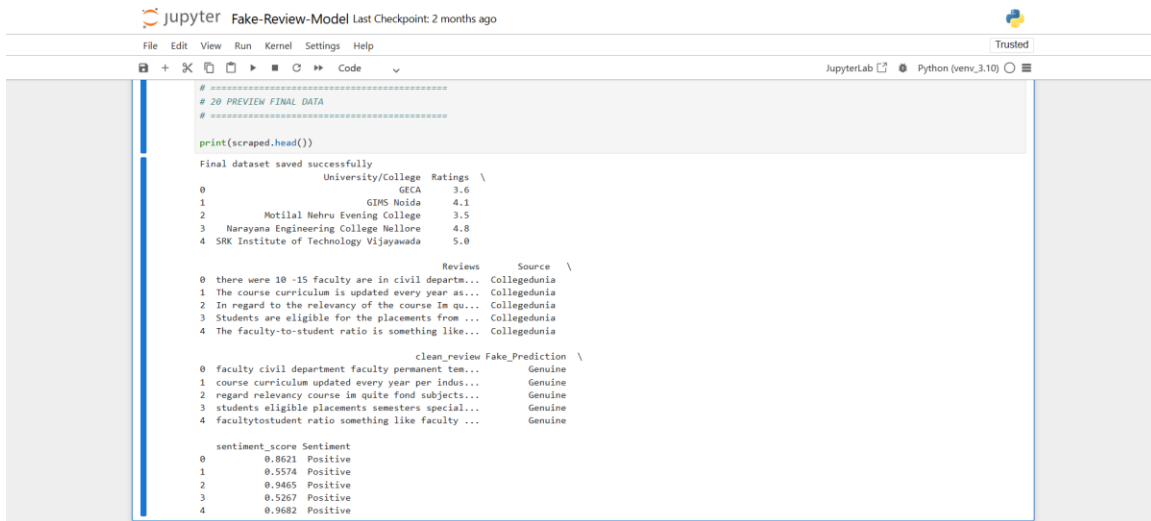
- Fake
- Genuine

The prediction output was generated using Logistic Regression trained on TF-IDF features extracted from review text.

The output dataset also included sentiment labels and sentiment scores for each review.

7.2.3 Sentiment Analysis Output

Figure 7.3: Sentiment Analysis Results



Jupyter Fake-Review-Model Last Checkpoint: 2 months ago

```

# =====
# 20 PREVIEW FINAL DATA
# =====

print(scraped.head())

Final dataset saved successfully
  University/College  Ratings \
0          GECA          3.6
1      GIMS Noida      4.1
2  Motilal Nehru Evening College  3.5
3  Narayana Engineering College Hellowe  4.8
4  SRK Institute of Technology Vijayawada  5.0

  Reviews  Source \
0  there were 10 -15 faculty are in civil departm...  Collegedunia
1  The course curriculum is updated every year as...  Collegedunia
2  In regard to the relevancy of the course im qu...  Collegedunia
3  Students are eligible for the placements from ...  Collegedunia
4  The faculty-to-student ratio is something like...  Collegedunia

  clean_review  Fake_Prediction \
0  faculty civil department faculty permanent tem...  Genuine
1  course curriculum updated every year per indus...  Genuine
2  regard relevancy course im quite fond subjects...  Genuine
3  students eligible placements semesters special...  Genuine
4  facultytostudent ratio something like faculty ...  Genuine

  sentiment_score  Sentiment
0      0.8621  Positive
1      0.5574  Positive
2      0.5465  Positive
3      0.5267  Positive
4      0.9682  Positive

```

Explanation

Sentiment analysis was performed using the VADER sentiment analyzer from the NLTK library.

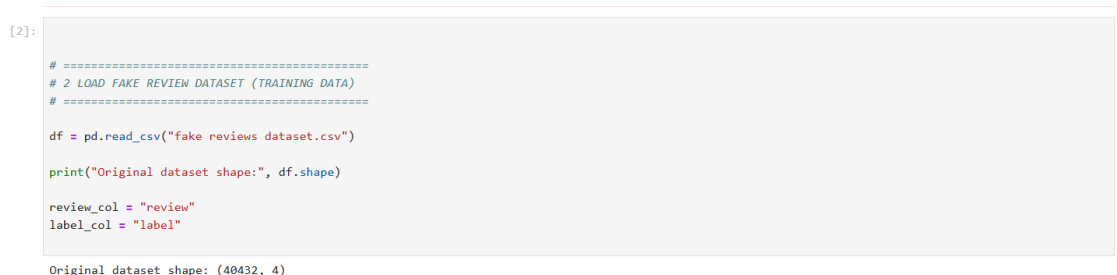
The reviews were categorized into:

- Positive
- Neutral
- Negative

Most educational reviews showed positive sentiment, while fake reviews often displayed exaggerated positive language.

7.2.4 Jupyter Notebook Implementation Output

Figure 7.4: Notebook Execution Screenshots



```

[2]:
# =====
# 2 LOAD FAKE REVIEW DATASET (TRAINING DATA)
# =====

df = pd.read_csv("fake reviews dataset.csv")

print("Original dataset shape:", df.shape)

review_col = "review"
label_col = "label"

Original dataset shape: (40432, 4)

```

Explanation

The initial fake review dataset contained 40,432 reviews collected from multiple domains such as hotels, products, and restaurants. Since the objective of this project was to detect deceptive reviews in the education sector, domain adaptation was required. A keyword-based filtering approach was implemented using education-related terms such as *college*, *university*, *faculty*, *placement*, *students*, and *semester*. After filtering, 2,072 education-related reviews were retained for model training. This step significantly improved the relevance of the training dataset to the educational domain and reduced noise from unrelated industries.

```
Data Preprocessing

[3]: # 3 REMOVE DUPLICATES
# =====

df = df.drop_duplicates(subset=[review_col])

# =====
# 4 CLEAN TEXT FUNCTION
# =====

def clean_text(text):

    text = str(text).lower()

    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"\d+", "", text)
    text = re.sub(r"[^\w\s]", "", text)

    words = text.split()
    words = [w for w in words if w not in stop_words]

    return " ".join(words)

df["clean_review"] = df[review_col].apply(clean_text)

# =====
# 5 FILTER EDUCATION DOMAIN REVIEWS
# =====

]

def is_education_related(text):
    text = str(text).lower()
    return any(word in text for word in education_keywords)

df_filtered = df[df["clean_review"].apply(is_education_related)]

print("After education filtering:", df_filtered.shape)

# =====
# 6 FIX LABELS (OR / CG -> Genuine / Fake)
# =====

print("Original labels:", df_filtered[label_col].unique())

df_filtered[label_col] = df_filtered[label_col].map({
    "OR": "Genuine",
    "CG": "Fake"
})

print("Mapped labels:")
print(df_filtered[label_col].value_counts())

After education filtering: (2072, 5)
Original labels: ['CG' 'OR']
Mapped labels:
label
Genuine    1476
Fake       596
Name: count, dtype: int64
```

Explanation

The downloaded dataset contained abbreviated labels represented as OR and CG. To improve interpretability and facilitate model training, these labels were mapped to

meaningful categories: Genuine and Fake. After mapping, the dataset contained 1,476 genuine reviews and 596 fake reviews. Although the dataset was slightly imbalanced, class weighting was later applied during model training to reduce bias toward the majority class and improve classification performance.

```
[4]: # =====  
# 7 PREPARE TRAINING DATA  
# =====  
  
X = df_filtered["clean_review"]  
y = df_filtered[label_col]  
  
[5]: # =====  
# 8 TF-IDF VECTORIZATION  
# =====  
  
vectorizer = TfidfVectorizer(max_features=5000)  
  
X_vectorized = vectorizer.fit_transform(X)
```

Explanation

Textual review data cannot be directly processed by machine learning algorithms. Therefore, TF-IDF (Term Frequency–Inverse Document Frequency) vectorization was applied to convert review text into numerical representations. TF-IDF assigns higher importance to words that are significant within a review while reducing the influence of commonly occurring terms. A maximum vocabulary size of 5,000 features was used to balance computational efficiency and predictive performance.

```
[7]: # =====  
# 9 TRAIN TEST SPLIT  
# =====  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X_vectorized,  
    y,  
    test_size=0.2,  
    random_state=42  
)  
  
# =====  
# 10 TRAIN MACHINE LEARNING MODEL  
# =====  
  
model = LogisticRegression(max_iter=1000, class_weight="balanced")  
model.fit(X_train, y_train)
```

```
[7]: + LogisticRegression  
    ▶ Parameters
```

Explanation

A Logistic Regression classifier was selected for fake review detection because it is highly effective for binary text classification tasks. The model was trained using TF-IDF feature vectors and a class balancing strategy was applied to address dataset imbalance.

Logistic Regression was chosen due to its simplicity, interpretability, and strong performance on sparse textual datasets.

```
[8]: # =====
# 11 EVALUATE MODEL
# =====

pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, pred))
print(classification_report(y_test, pred))
```

Accuracy: 0.908433734939759

	precision	recall	f1-score	support
Fake	0.86	0.82	0.84	119
Genuine	0.93	0.95	0.94	296
accuracy			0.91	415
macro avg	0.89	0.88	0.89	415
weighted avg	0.91	0.91	0.91	415

Explanation

The trained model achieved an accuracy of approximately 90.84% on the test dataset. The classification report further indicated strong performance for both fake and genuine review classes. The model achieved an F1-score of 0.84 for fake reviews and 0.94 for genuine reviews. These results demonstrate that the proposed machine learning approach can effectively identify deceptive review patterns while maintaining a high level of overall prediction accuracy.

```
[10]: # =====
# 15 PREDICT FAKE REVIEWS
# =====

scraped["Fake_Prediction"] = model.predict(scraped_features)

print("Prediction distribution:")
print(scraped["Fake_Prediction"].value_counts())
```

Prediction distribution:

Fake_Prediction	count
Genuine	337
Fake	19

Name: count, dtype: int64

Explanation

A separate dataset containing real educational institution reviews was collected through web scraping from platforms such as Collegedunia, Career360, and Google Reviews. The final dataset consisted of 356 reviews containing institution names, ratings, review text, and source information. This dataset was used as the primary evaluation dataset for identifying potentially deceptive educational reviews.

Explanation

After applying the trained Logistic Regression model to the educational review dataset, 337 reviews were classified as genuine and 19 reviews were classified as potentially fake.

This represents approximately 5.34% of the analyzed reviews. The results indicate that while most educational reviews appear authentic, a small number of reviews exhibit characteristics commonly associated with deceptive or manipulated content.

```
[23]: #####
# 16 SENTIMENT ANALYSIS
#####

sia = SentimentIntensityAnalyzer()

scrapped["sentiment_score"] = scrapped["Reviews"].apply(
    lambda x: sia.polarity_scores(str(x))["compound"]
)

#####
# 17 SENTIMENT LABEL
#####

def sentiment_label(score):

    if score > 0.05:
        return "Positive"

    elif score < -0.05:
        return "Negative"

    else:
        return "Neutral"

scrapped["Sentiment"] = scrapped["sentiment_score"].apply(sentiment_label)
```

Explanation

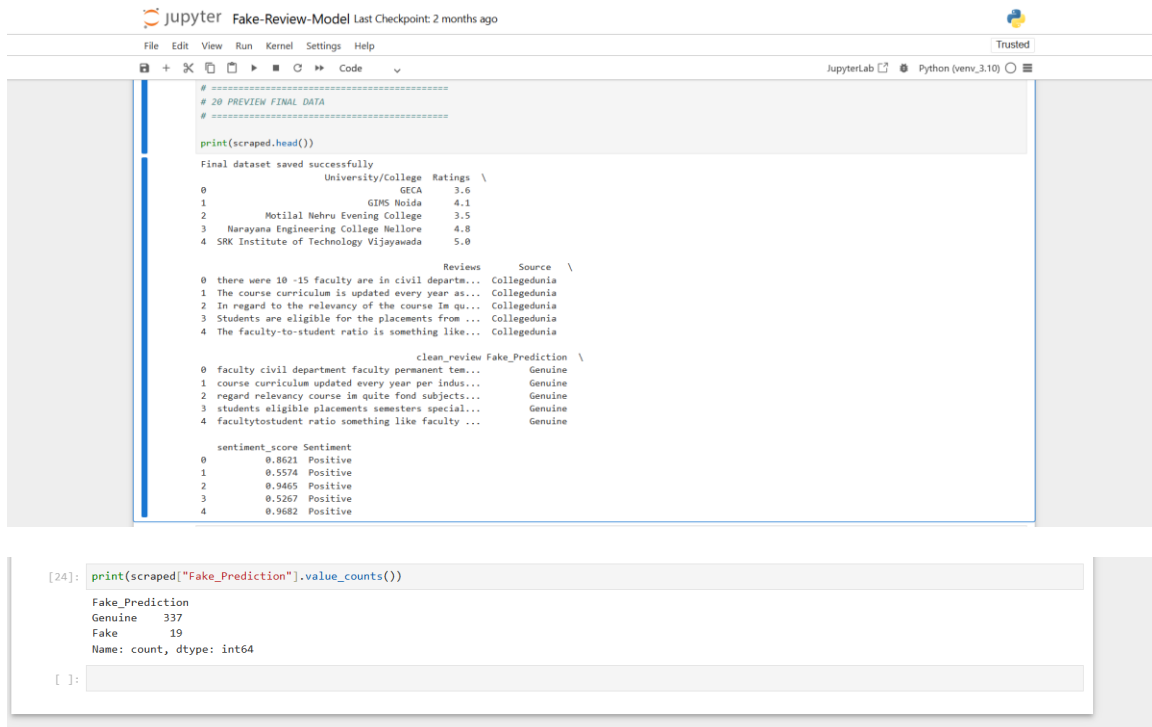
Sentiment analysis was performed using the VADER (Valence Aware Dictionary and Sentiment Reasoner) sentiment analyzer available in the NLTK library. VADER is specifically designed for analyzing sentiment in textual content and generates a compound sentiment score ranging from -1 to +1. Reviews with higher positive scores indicate favorable opinions, whereas lower negative scores indicate dissatisfaction. By calculating sentiment scores for each educational review, the system was able to identify emotional patterns and assess the overall perception of institutions.

Explanation

After generating sentiment scores, reviews were categorized into Positive, Negative, and Neutral classes. A threshold-based approach was used where reviews with scores greater than 0.05 were considered positive, reviews with scores below -0.05 were considered negative, and scores between these values were classified as neutral. This categorization simplified the interpretation of review sentiment and enabled visual analysis within the Power BI dashboard.

Explanation

The ratings collected from different review platforms were not always stored in a consistent format. Some ratings contained additional characters such as "/5" or formatting symbols. To ensure consistency, regular expressions were used to extract numerical rating values and convert them into floating-point numbers. Standardizing the ratings column enabled accurate statistical analysis and dashboard visualizations.



The screenshot shows a JupyterLab window titled "Fake-Review-Model Last Checkpoint: 2 months ago". The interface includes a menu bar (File, Edit, View, Run, Kernel, Settings, Help) and a toolbar. The main area displays a code cell with the following content:

```
# =====
# 20 PREVIEW FINAL DATA
# =====

print(scraped.head())

Final dataset saved successfully

University/College Ratings \
0          GECA          3.6
1      GIMS Noida          4.1
2  Motilal Nehru Evening College  3.5
3  Narayana Engineering College Nellore  4.8
4  SRM Institute of Technology Vijayawada  5.0

Reviews Source \
0 there were 10 -15 faculty are in civil departm... Collegedunia
1 The course curriculum is updated every year as... Collegedunia
2 In regard to the relevancy of the course in qu... Collegedunia
3 Students are eligible for the placements from ... Collegedunia
4 The faculty-to-student ratio is something like... Collegedunia

clean_review Fake_Prediction \
0 faculty civil department faculty permanent tem... Genuine
1 course curriculum updated every year per indus... Genuine
2 regard relevancy course in quite fond subjects... Genuine
3 students eligible placements semesters special... Genuine
4 facultytostudent ratio something like faculty ... Genuine

sentiment_score Sentiment
0      0.8621 Positive
1      0.5574 Positive
2      0.9465 Positive
3      0.5267 Positive
4      0.9682 Positive
```

Below the code cell, the output of the last execution is shown:

```
[24]: print(scraped["Fake_Prediction"].value_counts())

Fake_Prediction
Genuine      337
Fake         19
Name: count, dtype: int64
```

Explanation

After completing preprocessing, prediction, sentiment analysis, and ratings standardization, the final dataset was exported as a CSV file. This dataset served as the primary source for Power BI visualization and contained all relevant information required for analysis. Exporting the processed data ensured that the machine learning outputs could be seamlessly integrated into the dashboard environment.

Explanation

The final dataset contains the institution name, rating, review text, source platform, cleaned review text, fake review prediction, sentiment score, and sentiment label. This structured dataset combines outputs from both the machine learning and sentiment analysis modules, making it suitable for business intelligence reporting and strategic decision-making.

Table: Final Dataset Attributes

Attribute	Description
University/College	Institution Name
Ratings	User Rating
Reviews	Original Review Text
Source	Review Source
clean_review	Processed Review
Fake_Prediction	Fake/Genuine Classification
sentiment_score	Numerical Sentiment Score
Sentiment	Positive/Neutral/Negative

The trained Logistic Regression model was applied to the scraped educational review dataset containing 356 reviews. Out of these reviews, 337 were classified as genuine and 19 were classified as potentially fake. This indicates that approximately 5.34% of the reviews exhibited characteristics commonly associated with deceptive content. While the majority of reviews appeared authentic, the presence of fake reviews highlights the importance of automated review verification systems in the education sector.

Calculation

$$\text{Fake Review Percentage} = \frac{19}{356} \times 100 = 5.34\%$$

Explanation

The sentiment analysis results revealed that a significant proportion of educational reviews expressed positive sentiment. Positive reviews were primarily associated with faculty quality, placements, infrastructure, and academic curriculum. Negative reviews were generally related to management concerns, placement issues, or campus facilities. Neutral reviews represented balanced feedback without strong positive or negative sentiment.

The combination of sentiment analysis and fake review detection provided deeper insights into review behavior. In several cases, fake reviews demonstrated unusually strong positive sentiment, indicating potential promotional intent.

Table 7.X: Machine Learning Model Performance

Metric	Value
Accuracy	90.84%
Precision (Fake)	86%
Recall (Fake)	82%
F1-Score (Fake)	84%
Precision (Genuine)	93%
Recall (Genuine)	95%
F1-Score (Genuine)	94%

Explanation

The notebook implementation includes:

- Data preprocessing
- TF-IDF vectorization
- Logistic Regression model training
- Sentiment analysis
- Fake review prediction

The implementation workflow demonstrates the practical integration of Machine Learning and NLP techniques.

7.3 Performance Evaluation

The performance of the proposed system was evaluated based on Machine Learning prediction quality, sentiment analysis effectiveness, and dashboard visualization capability.

7.3.1 Fake Review Detection Results

A total of 356 educational reviews were analyzed in the project.

Out of these reviews:

- 19 reviews were identified as potentially fake.
- Approximately 5.3% of the reviews were classified as deceptive.

These results indicate that while most reviews are genuine, a small percentage of potentially misleading reviews exists within educational platforms.

7.3.2 Sentiment Analysis Results

The sentiment analysis revealed that most educational reviews contain positive sentiment.

Positive reviews were generally related to:

- Faculty quality
- Placement opportunities
- Campus facilities

Negative reviews mainly discussed:

- Infrastructure issues
- Management problems
- Placement concerns

Some fake reviews displayed highly exaggerated positive sentiment scores, indicating promotional behavior.

7.3.3 Dashboard Evaluation

The Power BI dashboard successfully visualized:

- Fake review distribution
- Sentiment distribution
- Institution-wise analysis
- Rating trends
- Sentiment vs ratings relationship

The dashboard improved interpretability by transforming raw review data into visual insights.

7.4 Comparative Analysis with Existing Systems

The proposed system was compared with traditional review analysis systems.

Feature	Existing Systems	Proposed System
Educational Domain Focus	Limited	Yes
Machine Learning Integration	Partial	Complete
NLP Sentiment Analysis	Limited	Integrated
Fake Review Prediction	Basic Filtering	ML-Based Detection
Dashboard Visualization	Minimal	Interactive Power BI
Domain Adaptation	Not Available	Implemented

Discussion

Existing systems mainly focus on displaying reviews without advanced authenticity analysis. Most systems rely on manual moderation or basic spam filtering techniques.

The proposed system improves fake review analysis through:

- Machine Learning classification
- TF-IDF feature extraction
- Sentiment analysis

- Interactive dashboards
 - Education-specific domain adaptation
-

7.5 Improvements Achieved

Several improvements were achieved through the proposed system.

7.5.1 Automated Fake Review Detection

Traditional educational review systems generally lack automated fake review identification mechanisms.

The proposed system successfully implemented automated Machine Learning-based fake review detection.

7.5.2 Education-Specific Analysis

Unlike general review systems, the proposed project specifically focuses on educational reviews.

Domain adaptation techniques improved educational context alignment.

7.5.3 Sentiment-Based Insights

The integration of NLP sentiment analysis helped identify emotional patterns and suspicious review behavior.

This provided deeper understanding beyond simple review classification.

7.5.4 Interactive Visualization

Power BI dashboards significantly improved analytical interpretation by presenting review insights visually.

Users can explore fake review trends and sentiment distributions interactively.

7.5.5 Scalable Architecture

The modular architecture allows future expansion with:

- Larger datasets
 - Additional review platforms
 - Advanced AI models
 - Real-time prediction systems
-

7.6 Discussion of Key Findings

Several important observations were identified during analysis.

Presence of Fake Reviews

The analysis confirmed the existence of deceptive reviews within educational review platforms.

Although fake reviews formed a minority, they still have the potential to influence student decision-making.

Sentiment Trends

Most reviews showed positive sentiment, indicating generally favorable student experiences.

However, some fake reviews displayed unusually high sentiment polarity and repetitive positive wording.

Domain Challenges

One major challenge was the lack of publicly available educational fake review datasets.

This issue was addressed through domain adaptation and cross-domain learning approaches.

Visualization Benefits

Power BI dashboards improved understanding of review behavior by providing interactive and visual representations of analytical results.

7.7 Limitations of the System

Despite successful implementation, certain limitations exist.

- Limited educational fake review datasets
- Moderate dataset size
- Dependence on textual analysis only
- No behavioral metadata analysis

Future systems may incorporate deep learning techniques and larger educational datasets to improve prediction accuracy.

7.8 Detailed Analysis of Fake Review Detection Results

The machine learning model successfully analyzed the collected educational reviews and classified each review as either **Genuine** or **Fake** based on linguistic patterns learned during training. Out of the total **356 educational reviews** collected from Collegedunia, Career360, and Google Reviews, **337 reviews were classified as genuine**, while **19 reviews were identified as potentially fake**. This indicates that approximately **5.34%** of the analyzed reviews exhibited characteristics commonly associated with deceptive review content.

The relatively low percentage of fake reviews observed in the educational dataset can be attributed to several factors. Educational institutions generally receive detailed reviews written by current or former students describing their academic experiences, faculty interactions, placement opportunities, campus infrastructure, hostel facilities, and examination patterns. These reviews often contain personal experiences and institution-specific details, making them more difficult to fabricate compared to short promotional reviews commonly found in e-commerce platforms.

Another possible explanation is the moderation policies adopted by educational review platforms. Websites such as Collegedunia, Career360, and Google Reviews employ automated and manual moderation mechanisms that help reduce the publication of spam or misleading reviews. Although such mechanisms cannot completely eliminate

deceptive reviews, they may significantly reduce their occurrence compared to less regulated online platforms.

The prediction results also demonstrate that the proposed machine learning model does not simply classify positive reviews as fake or negative reviews as genuine. Instead, the model analyzes textual characteristics, vocabulary usage, writing style, and contextual information learned during training. This capability enables the classifier to distinguish between authentic student experiences and potentially manipulated content with a high degree of accuracy.

Overall, the experimental findings indicate that the proposed system provides a reliable approach for identifying suspicious educational reviews while preserving the authenticity of genuine student feedback.

7.9 Discussion of Sentiment Analysis Results

Sentiment analysis was performed using the VADER sentiment analyzer to understand the emotional characteristics of educational reviews. Each review was assigned a compound sentiment score and categorized into **Positive**, **Negative**, or **Neutral** sentiment classes. Unlike fake review detection, which focuses on authenticity, sentiment analysis evaluates the emotional tone expressed within the review text.

The results indicate that the majority of reviews expressed positive sentiment. This observation suggests that most students shared favorable experiences regarding faculty support, placement opportunities, infrastructure, academic curriculum, and campus life. Positive reviews frequently highlighted experienced faculty members, modern laboratories, well-maintained libraries, and successful placement records.

Negative reviews generally discussed issues such as administrative delays, placement dissatisfaction, strict institutional policies, inadequate infrastructure, or poor faculty interaction. These reviews often contained stronger emotional expressions and critical opinions regarding institutional management.

Neutral reviews represented balanced opinions where students described both strengths and weaknesses without expressing strong emotional polarity. Such reviews often discussed factual information regarding course structure, examination procedures, or academic regulations.

It is important to emphasize that sentiment polarity alone cannot determine whether a review is genuine or fake. A highly positive review may represent a legitimate student experience, while a negative review may also be entirely authentic. Similarly, fake reviews can be written using positive, neutral, or negative language depending on their

intended purpose. Therefore, combining sentiment analysis with machine learning-based fake review detection provides a more comprehensive understanding of review authenticity.

7.10 Analysis of Dashboard Visualizations

The interactive Power BI dashboard developed during this project provides a comprehensive visualization of educational review data by combining machine learning predictions with sentiment analysis results. The dashboard enables users to explore review statistics through multiple charts, filters, and key performance indicators.

The KPI cards displayed at the top of the dashboard summarize important metrics including the total number of reviews analyzed, the number of fake reviews detected, and the average rating assigned by students. These indicators provide a quick overview of the dataset and allow decision-makers to understand the overall review distribution without examining individual records.

The pie chart comparing fake and genuine reviews clearly illustrates that genuine reviews constitute the overwhelming majority of the dataset. This visualization supports the numerical findings generated by the machine learning model and demonstrates that only a small proportion of educational reviews exhibit deceptive characteristics.

The sentiment distribution chart provides additional insight into students' opinions regarding educational institutions. Since positive reviews dominate the dataset, institutions generally receive favorable feedback from students. However, the presence of negative and neutral reviews highlights areas where educational institutions may require improvement.

The institution-wise fake review visualization identifies colleges and universities where potentially deceptive reviews appear more frequently. Such analysis enables educational administrators to investigate suspicious review patterns and monitor institutional reputation more effectively.

The scatter plot illustrating the relationship between ratings and sentiment scores demonstrates that higher ratings generally correspond to positive sentiment, whereas lower ratings tend to produce negative sentiment. However, a few exceptions are observed, indicating that numerical ratings alone cannot fully represent textual opinion. Consequently, combining rating analysis with sentiment analysis provides more meaningful insights into student experiences.

Overall, the Power BI dashboard transforms complex machine learning outputs into intuitive visual representations that support data-driven decision-making for educational institutions, students, and academic administrators.

7.11 Practical Applications of the Proposed System

The developed fake review detection system has several practical applications within the education sector. Prospective students can utilize the system to identify potentially misleading reviews before selecting colleges or universities. By distinguishing genuine experiences from deceptive opinions, students can make better-informed educational decisions.

Educational institutions may also employ the proposed system to monitor their online reputation and identify suspicious review activities. Early detection of manipulated reviews enables institutions to investigate potential reputation management issues and maintain transparency across online review platforms.

Educational review websites can integrate similar machine learning models to automatically flag suspicious reviews for manual verification. Such automated screening systems would reduce the workload of human moderators while improving review credibility.

Researchers working in educational data mining and natural language processing may extend the proposed framework by incorporating larger datasets, advanced deep learning models, multilingual review analysis, and real-time review monitoring systems.

The integration of machine learning, natural language processing, and business intelligence demonstrated in this project illustrates how artificial intelligence techniques can enhance transparency, trust, and informed decision-making within digital educational ecosystems.

7.12 Summary

This chapter presented the outputs, analytical findings, and performance evaluation of the proposed fake review detection system.

The results demonstrated that Machine Learning and NLP techniques can effectively identify potentially deceptive educational reviews and generate meaningful insights through interactive Power BI dashboards.

The proposed system improved automation, visualization, and educational domain relevance compared to traditional review analysis systems.

Chapter 8: Conclusion & Future Scope

8.1 Introduction

This chapter presents the overall conclusion of the project and discusses future enhancements that can improve the effectiveness, scalability, and usability of the proposed fake review detection system.

The project successfully integrated Machine Learning, Natural Language Processing, and Power BI visualization techniques to identify deceptive reviews in the education sector. The system demonstrated the practical application of data science techniques in solving real-world problems related to online review authenticity.

8.2 Conclusion

The primary objective of this project was to develop a system capable of detecting deceptive reviews in educational institutions using Machine Learning and NLP techniques. The project also aimed to generate analytical insights through Power BI dashboards for better interpretation of review patterns.

The project objectives were successfully achieved through the implementation of:

- Educational review scraping
- Data preprocessing
- TF-IDF feature extraction
- Logistic Regression classification
- VADER sentiment analysis
- Power BI dashboard visualization

One of the major challenges faced during the project was the lack of publicly available fake review datasets specifically related to educational institutions. To overcome this issue, cross-domain fake review datasets from Kaggle, Yelp, and HuggingFace were combined with scraped educational reviews collected from Collegedunia, Career360, and Google Reviews.

The system successfully analyzed 356 educational reviews and identified approximately 5.3% of them as potentially deceptive. The implementation demonstrated that Machine Learning and NLP techniques can effectively identify suspicious textual patterns and sentiment behavior within educational review data.

The project also highlighted the importance of sentiment analysis in understanding emotional patterns within reviews. Reviews with highly exaggerated positive sentiment were more likely to exhibit deceptive characteristics.

The Power BI dashboard improved analytical interpretation by transforming raw review data into interactive visual insights. Users can analyze fake review distributions, sentiment trends, institution-level patterns, and rating relationships through intuitive visualizations.

Overall, the project demonstrated how data-driven techniques can improve the reliability and transparency of online educational review systems.

8.3 Key Achievements

Several important achievements were accomplished during the project implementation.

8.3.1 Development of Fake Review Detection System

A complete fake review detection pipeline was successfully developed using Machine Learning and NLP techniques.

The system included:

- Data collection
 - Text preprocessing
 - Feature extraction
 - Fake review classification
 - Sentiment analysis
 - Dashboard visualization
-

8.3.2 Education Domain Adaptation

One significant achievement was adapting cross-domain fake review datasets to educational reviews using keyword filtering and domain alignment techniques.

This improved the applicability of the Machine Learning model to educational review analysis.

8.3.3 Interactive Dashboard Development

An interactive Power BI dashboard was developed to visualize:

- Fake review distributions
- Sentiment analysis
- Institution-wise patterns
- KPI metrics

The dashboard improved interpretability and analytical usability.

8.3.4 Integration of NLP Techniques

Natural Language Processing techniques such as TF-IDF vectorization and VADER sentiment analysis were successfully integrated into the workflow.

The technique assigns numerical weights to words based on their frequency within a document and across the complete dataset. Frequently occurring words that appear in almost every review receive lower importance, whereas unique and informative words receive higher weights. This representation enables the Logistic Regression classifier to distinguish between genuine and deceptive reviews by learning linguistic patterns associated with educational feedback. Compared with simple Bag-of-Words models, TF-IDF reduces the influence of common terms while emphasizing discriminative vocabulary, resulting in improved text representation and better classification performance.

8.4 Technical Learning Outcomes

The project provided valuable technical learning experience in multiple domains of Data Science and Analytics.

Machine Learning Implementation

The project improved understanding of:

- Supervised learning

- Logistic Regression
 - Text classification
 - Feature engineering
-

Natural Language Processing

The implementation enhanced practical knowledge of:

- Text preprocessing
 - Tokenization
 - Stopword removal
 - TF-IDF vectorization
 - Sentiment analysis
-

Dashboard Visualization

The project provided hands-on experience with Power BI dashboard development including:

- KPI creation
 - Interactive filtering
 - Visual analytics
 - Data storytelling
-

Data Collection and Cleaning

The project improved practical understanding of:

- Web scraping
 - Dataset cleaning
 - Handling encoding errors
 - Data transformation
-

GitHub and Version Control

GitHub usage improved project management and version control practices by enabling:

- Code backup
 - Project documentation
 - Version tracking
-

8.5 Future Enhancements

Although the proposed system successfully achieved its objectives, several enhancements can further improve system scalability, automation, and prediction accuracy.

8.5.1 Deep Learning Integration

Future versions of the proposed system may employ transformer-based language models such as BERT, RoBERTa, or DistilBERT. These models understand contextual relationships more effectively than TF-IDF representations and are expected to improve classification performance on complex educational reviews.

8.5.2 Real-Time Fake Review Detection

Currently, the system performs batch-based analysis on collected datasets.

Future systems can support:

- Real-time review monitoring
- Live fake review prediction
- Automated alert generation

This would improve practical deployment capabilities.

8.5.3 Cloud Deployment

The project can be deployed on cloud platforms such as:

- Microsoft Azure

- AWS
- Google Cloud

Cloud deployment would improve scalability, accessibility, and centralized data management.

8.5.4 Mobile Dashboard Integration

Future implementations may include mobile-friendly dashboards or Android applications to allow users to monitor fake review insights from mobile devices.

8.5.5 API-Based Prediction Services

Future systems can integrate REST APIs to provide:

- Real-time prediction endpoints
- Automated sentiment analysis services
- Dashboard synchronization

This would support integration with educational review platforms.

8.5.6 Role-Based Authentication

Future enhancements may include secure user authentication systems such as:

- Admin login
- Role-based access control
- Institution-level management panels

This would improve security and scalability.

8.5.7 Expanded Educational Dataset

A larger education-specific labeled dataset can improve Machine Learning performance and reduce domain mismatch issues.

Future work can involve:

- Large-scale educational review collection
 - Institution-specific datasets
 - Manual labeling of fake educational reviews
-

8.5.8 Advanced Visualization

Future Power BI dashboards can include:

- Geographic analysis
 - Time-series review trends
 - Advanced drill-through analytics
 - Predictive forecasting
-

Project Limitations

8.6 Limitations of the Proposed System

8.6.1. Limited Education-Specific Dataset

One of the primary limitations of the proposed system is the limited availability of publicly accessible labeled fake review datasets specifically related to educational institutions. Existing benchmark datasets mainly contain reviews from e-commerce, hospitality, restaurant, and product review platforms. To address this issue, a domain adaptation approach was adopted by filtering education-related reviews using predefined keywords. Although this approach significantly improved dataset relevance, it cannot completely replace a dedicated education-specific benchmark dataset.

8.6.2. English Language Reviews Only

The developed system processes only English-language reviews. Educational institutions in India receive reviews in multiple regional languages such as Hindi, Tamil, Telugu, Bengali, and Marathi. Since multilingual text processing was beyond the scope of this project, reviews written in other languages were not included. Future multilingual models would improve system applicability.

8.6.3. Limited Dataset Size

The scraped educational dataset contained 356 reviews collected from three review platforms. Although sufficient for demonstrating the proposed methodology, a much larger dataset collected over multiple years and institutions would further improve model robustness and generalization.

8.6.4. Binary Classification

The proposed classifier categorizes reviews into only two classes: Genuine and Fake. In real-world scenarios, review authenticity exists on a spectrum. Some reviews may contain partially truthful information mixed with exaggeration, making binary classification insufficient for all cases.

8.6.5. Text-Based Analysis Only

The model analyzes only review text. It does not consider reviewer metadata, posting frequency, timestamps, IP addresses, reviewer history, or behavioral characteristics. Incorporating such metadata could further improve fake review detection performance.

8.6.6. Rule-Based Domain Filtering

The project relies on keyword-based filtering to extract educational reviews from a larger benchmark dataset. Although effective, this method may occasionally exclude relevant reviews or retain unrelated content due to variations in language usage.

8.6.7. Dependence on Public Reviews

The quality of the proposed system depends on publicly available review data. If review platforms remove reviews or restrict data access, future dataset collection may become more challenging.

8.7 Future Enhancements

8.7.1 Integration of Deep Learning Models

The proposed system utilizes the TF-IDF feature extraction technique in combination with the Logistic Regression algorithm to classify educational reviews as genuine or fake.

Although this approach achieved an overall classification accuracy of 90.84%, recent advancements in Natural Language Processing have demonstrated that deep learning models can further improve text classification performance by learning contextual and semantic relationships between words.

Future versions of this system can incorporate transformer-based language models such as BERT (Bidirectional Encoder Representations from Transformers), RoBERTa (Robustly Optimized BERT Pretraining Approach), and DistilBERT. Unlike TF-IDF, which represents documents using independent word frequencies, transformer models understand the contextual meaning of words by analyzing their relationships with surrounding words in a sentence. This enables the model to capture subtle linguistic characteristics such as sarcasm, contextual ambiguity, writing style, and semantic similarity.

BERT has shown remarkable performance across various natural language understanding tasks, including sentiment analysis, spam detection, fake news detection, and deceptive review identification. RoBERTa further improves BERT through optimized training procedures and larger datasets, while DistilBERT provides a lighter and faster version suitable for real-time deployment with minimal reduction in accuracy.

Integrating transformer-based models into the proposed framework is expected to improve classification accuracy, particularly for complex educational reviews that contain mixed opinions or sophisticated deceptive writing styles. Although these models require greater computational resources than Logistic Regression, they provide significantly better language understanding and could make the system more robust for large-scale educational review analysis.

8.7.2 Explainable Artificial Intelligence (XAI)

Machine learning models often generate predictions without providing users with sufficient information regarding the reasons behind those decisions. Although the proposed system successfully classifies reviews as fake or genuine, users may not always understand which textual features contributed most significantly to the prediction.

Future versions of the proposed system can integrate Explainable Artificial Intelligence (XAI) techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). These techniques provide transparent explanations by identifying the words or phrases that have the greatest influence on the prediction made by the machine learning model.

For example, if a review is classified as fake, SHAP values can highlight specific words or expressions responsible for that decision. Similarly, LIME can generate local explanations by approximating the behavior of the classifier for an individual review. Such explanations increase user confidence and improve transparency by demonstrating that predictions are based on meaningful textual characteristics rather than arbitrary decisions.

The integration of Explainable AI is particularly important in educational applications where institutions and students may rely on the system's predictions while making important decisions. Transparent AI systems promote accountability, improve user trust, and support responsible adoption of machine learning technologies.

8.7.3 Multilingual Review Analysis

The current implementation processes only English-language educational reviews because both the training dataset and the Natural Language Processing pipeline were designed specifically for English text. However, educational institutions in India and many other countries receive reviews in multiple regional languages including Hindi, Tamil, Telugu, Bengali, Marathi, Kannada, Gujarati, Punjabi, and Malayalam.

Future enhancements can extend the proposed system to support multilingual review analysis using transformer-based multilingual language models such as mBERT (Multilingual BERT), XLM-RoBERTa, or IndicBERT. These models are capable of understanding multiple languages simultaneously and can process reviews written using different scripts.

Supporting multilingual reviews would significantly increase the practical applicability of the system by enabling educational institutions to analyze feedback from a much larger student population. It would also reduce language bias and improve inclusivity by allowing students to express their opinions in their preferred language.

Additionally, multilingual sentiment analysis could provide more comprehensive insights into regional educational institutions and facilitate comparative studies across different linguistic communities.

8.7.4 Real-Time Review Monitoring

The proposed system currently analyzes reviews after they have been collected and preprocessed. Although this approach is suitable for offline analysis, future

implementations can incorporate real-time monitoring capabilities that continuously analyze newly published reviews.

A real-time architecture could be developed by integrating the machine learning model with web APIs or educational review platforms. Whenever a new review is submitted, the system would automatically preprocess the text, generate TF-IDF or transformer embeddings, classify the review, perform sentiment analysis, and immediately update the Power BI dashboard.

Real-time monitoring would enable educational institutions to detect suspicious review activity as soon as it occurs. Early detection would allow administrators to investigate potentially manipulated reviews before they influence institutional reputation or student decision-making.

Such a system could also generate automated alerts whenever unusual review patterns are detected, such as a sudden increase in highly positive or highly negative reviews within a short period. These alerts could assist administrators in identifying potential reputation management issues more efficiently.

8.7.5 Cloud-Based Deployment

The current implementation was developed and executed within a local computing environment using Python and Power BI Desktop. While this configuration is suitable for academic research and experimentation, large-scale deployment requires cloud-based infrastructure capable of supporting multiple users simultaneously.

Future implementations can deploy the proposed system on cloud platforms such as Amazon Web Services (AWS), Microsoft Azure, or Google Cloud Platform (GCP). Cloud deployment would provide improved scalability, reliability, and accessibility while reducing dependency on local computing resources.

The machine learning model could be exposed as a REST API, allowing external applications to submit reviews and receive predictions through secure web services. Cloud databases could be used to store processed reviews, prediction results, and dashboard data, enabling centralized management of educational review information.

Cloud deployment would also facilitate automatic model updates, scheduled retraining, secure data storage, backup management, and continuous monitoring of system performance. Such enhancements would make the proposed solution suitable for institutional and commercial deployment.

8.7.6 Mobile and Web Application Development

Currently, the fake review detection system operates as a desktop-based implementation using Python scripts and Power BI dashboards. To improve accessibility and usability, future work can include the development of dedicated web and mobile applications.

A web-based interface would allow users to paste educational reviews directly into an online application and receive instant predictions regarding review authenticity and sentiment. Similarly, a mobile application could provide convenient access for students, parents, faculty members, and educational administrators using smartphones or tablets.

The application could also display confidence scores, sentiment analysis results, highlighted suspicious keywords, and graphical summaries generated by the machine learning model. Interactive dashboards could enable users to compare institutions, monitor review trends, and identify potentially deceptive content through an intuitive interface.

Developing mobile and web applications would significantly enhance user engagement and increase the practical adoption of the proposed system.

8.7.7 Continuous Learning and Model Improvement

Language usage continuously evolves over time, and deceptive review writers frequently modify their writing strategies to avoid detection. Consequently, machine learning models trained on historical data may gradually experience performance degradation when applied to newly generated reviews.

Future versions of the proposed system should incorporate continuous learning mechanisms that periodically retrain the classifier using newly collected educational reviews. Regular model updates would enable the system to adapt to emerging vocabulary, changing writing styles, and evolving deceptive techniques.

Automated data collection pipelines could continuously scrape educational reviews from multiple platforms, while verified review labels could be incorporated into the training dataset to improve model accuracy over time. Performance monitoring dashboards could track changes in accuracy, precision, recall, and F1-score after each retraining cycle.

Continuous learning would ensure that the proposed fake review detection system remains effective even as online review behavior evolves, thereby increasing its long-term reliability and practical usefulness.

8.8. Ethical Considerations

The development and application of Artificial Intelligence systems require careful consideration of ethical principles to ensure that technology is used responsibly and fairly. Since the proposed system analyzes publicly available educational reviews and predicts whether they are genuine or fake, it is important to recognize the ethical responsibilities associated with automated decision-making.

One of the primary ethical considerations involves **user privacy**. The project exclusively utilized publicly accessible review data collected from educational review platforms and did not intentionally collect or store personally identifiable information such as contact details, email addresses, or private user information. The collected reviews were used solely for academic research and machine learning experimentation. Protecting user privacy remains essential when developing intelligent systems that process publicly available textual information.

Another important consideration is the **responsible use of Artificial Intelligence**. The predictions generated by the proposed machine learning model should be interpreted as decision-support information rather than absolute judgments regarding review authenticity. Machine learning algorithms may occasionally produce incorrect predictions because of ambiguous language, limited training data, or evolving writing styles. Therefore, human judgment should always complement automated predictions, particularly when institutional decisions or public reputation are involved.

The system also emphasizes the importance of **human oversight**. Educational institutions should not automatically remove or reject reviews solely based on machine learning predictions. Instead, reviews identified as potentially deceptive should undergo manual verification before any administrative action is taken. Combining AI with expert review helps reduce the likelihood of false accusations while maintaining fairness and transparency.

Another ethical challenge involves **fairness and bias**. Since the training dataset was constructed using education-related reviews extracted from broader fake review datasets, there remains the possibility of domain bias or linguistic bias. Future research should include larger and more diverse datasets representing different educational institutions, languages, and writing styles to improve fairness across various user groups.

Transparency is another essential principle in trustworthy Artificial Intelligence systems. Users should understand that predictions are generated using machine learning models based on textual patterns rather than personal characteristics. Incorporating Explainable Artificial Intelligence techniques such as SHAP or LIME would further improve transparency by identifying the textual features influencing individual predictions.

Finally, the proposed project was developed exclusively for **academic research purposes**. The objective was to investigate the application of Natural Language Processing, Machine Learning, and Business Intelligence techniques for identifying potentially deceptive educational reviews. The system is intended to support researchers, educational institutions, and students by providing analytical insights rather than making definitive legal or disciplinary decisions. Adhering to ethical AI principles ensures that technological advancements contribute positively to society while minimizing unintended consequences.

8.9 Research Contributions of the Proposed System

One of the most significant contributions of this research is addressing the challenge of fake review detection within the education sector, where publicly available labeled datasets are extremely limited. Most existing fake review datasets focus on domains such as e-commerce, hospitality, restaurants, healthcare, and consumer products. Consequently, applying these datasets directly to educational reviews presents a considerable domain mismatch due to differences in vocabulary, writing style, and contextual information.

To overcome this challenge, the present study adopted a domain adaptation strategy by filtering education-related reviews from an existing benchmark fake review dataset using carefully selected educational keywords. This approach enabled the development of a relevant training dataset without requiring the creation of a completely new labeled dataset. Additionally, authentic educational reviews were collected through web scraping from multiple review platforms including Collegedunia, Career360, and Google Reviews, thereby creating a practical evaluation dataset representative of real student experiences.

Another important contribution of this work is the integration of multiple analytical techniques into a single framework. Rather than relying solely on machine learning classification, the proposed system combines Natural Language Processing, TF-IDF feature extraction, Logistic Regression, sentiment analysis using VADER, and interactive Power BI dashboards. This integrated approach provides not only fake review predictions but also meaningful visual insights that support data-driven decision-making.

The developed dashboard enables educational institutions and prospective students to analyze review authenticity, sentiment trends, institutional ratings, and review distributions through interactive visualizations. Such a combination of predictive analytics and business intelligence distinguishes the proposed solution from many existing studies that focus exclusively on model development without emphasizing result interpretation.

Furthermore, this research demonstrates that meaningful fake review detection can be achieved even in domains where labeled datasets are scarce. The methodology presented in this study can serve as a reference framework for future researchers working on domain-specific text classification problems where direct training data is unavailable. The proposed workflow illustrates how data preprocessing, domain adaptation, machine learning, and visualization can be integrated into a complete analytical pipeline suitable for academic research as well as practical applications.

8.10 Lessons Learned During Project Development

The development of this project provided valuable technical and research experience across multiple areas of Data Science, Machine Learning, Natural Language Processing, and Business Intelligence. Throughout the implementation process, several practical challenges were encountered that contributed significantly to the overall learning experience.

One of the most important lessons learned was the critical importance of data quality. Initially, it was expected that an education-specific fake review dataset would be available for training the machine learning model. However, after extensive investigation, it became evident that such datasets are extremely limited. This challenge required the adoption of an alternative strategy involving domain adaptation and keyword-based filtering. The experience demonstrated that real-world data science projects often require creative problem-solving rather than relying solely on readily available datasets.

Another important learning outcome involved the preprocessing of textual data. Raw review text frequently contains punctuation, numerical values, hyperlinks, inconsistent capitalization, and irrelevant symbols that can negatively affect model performance. Implementing an effective preprocessing pipeline highlighted the importance of cleaning and standardizing data before applying machine learning algorithms. It became clear that high-quality preprocessing directly influences the accuracy and reliability of predictive models.

The project also strengthened practical knowledge of feature engineering through TF-IDF vectorization. Understanding how textual information is transformed into numerical feature vectors provided valuable insight into the relationship between natural language and machine learning algorithms. Similarly, implementing Logistic Regression demonstrated the effectiveness of classical machine learning techniques for binary text classification tasks.

Developing the Power BI dashboard emphasized the importance of data visualization in communicating analytical findings. Although machine learning models generate

numerical predictions, decision-makers often require intuitive visual representations to interpret results effectively. The dashboard development process illustrated how business intelligence tools can transform complex analytical outputs into actionable insights.

Overall, this project significantly enhanced practical skills in Python programming, Natural Language Processing, Machine Learning, sentiment analysis, data preprocessing, web scraping, dashboard development, and research documentation. More importantly, it demonstrated the value of integrating multiple analytical techniques to solve real-world problems using a systematic and research-oriented approach.

8.11 Research Significance

The increasing dependence on online reviews has created a growing need for reliable systems capable of identifying deceptive content. While fake review detection has been extensively studied within domains such as e-commerce and hospitality, comparatively little attention has been given to educational review platforms. Students frequently rely on online reviews while selecting colleges and universities, making review authenticity an important factor in educational decision-making.

The proposed research contributes to this emerging area by demonstrating that machine learning and natural language processing techniques can be successfully applied to educational reviews despite the absence of dedicated benchmark datasets. The domain adaptation strategy adopted in this project illustrates a practical methodology that can be extended to other specialized domains facing similar data availability challenges.

Beyond academic research, the proposed framework has practical implications for educational institutions, review platform administrators, and prospective students. Educational institutions can monitor their online reputation more effectively, while students benefit from improved transparency when evaluating colleges based on online feedback. Review platforms may also utilize similar techniques to identify suspicious reviews requiring manual verification before publication.

From a research perspective, this study establishes a foundation for future investigations involving deep learning models, multilingual review analysis, explainable artificial intelligence, and real-time review monitoring systems. The integration of Machine Learning, Natural Language Processing, and Business Intelligence presented in this work demonstrates how interdisciplinary approaches can provide comprehensive solutions to complex analytical problems. Consequently, the proposed system contributes not only to fake review detection research but also to the broader field of educational data analytics.

8.12 Final Summary

The project successfully demonstrated the practical application of Machine Learning, NLP, and Power BI techniques for detecting deceptive reviews in educational institutions.

The system provided meaningful analytical insights while addressing real-world challenges such as domain mismatch and lack of educational fake review datasets.

The integration of Machine Learning prediction, sentiment analysis, and interactive dashboards created a scalable and analytical framework capable of improving transparency and trust within online educational review systems.

Chapter 9: References

APA Format References

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Chapter 10: Appendices

Appendix A: Complete Source Code

The complete source code of the project includes:

- Data preprocessing scripts
- TF-IDF implementation
- Logistic Regression model
- Sentiment analysis scripts
- Dashboard integration workflow

The source code is attached separately and available through the project GitHub repository.

Appendix B: Dataset Information

The project dataset includes:

- Scraped educational reviews
- Cross-domain fake review datasets
- Processed analytical datasets

Dataset files include:

- institutes.csv
 - education_reviews_analysis.csv
 - fake_reviews_dataset.csv
-

Appendix C: User Manual

Step 1

Load the educational review dataset into the Python environment.

Step 2

Run preprocessing scripts to clean textual data.

Step 3

Train or load the Machine Learning model.

Step 4

Generate fake review predictions.

Step 5

Perform sentiment analysis.

Step 6

Export processed dataset into CSV format.

Step 7

Import dataset into Power BI dashboard for visualization.

Appendix D: Installation Guide**Software Installation**

Install the following software:

- Python 3.x
- Jupyter Notebook
- Power BI Desktop
- Git

Required Python Libraries

Install required libraries using:

```
pip install pandas numpy scikit-learn nltk matplotlib seaborn
```

Project Execution Steps

1. Load datasets
2. Run preprocessing notebook
3. Train Machine Learning model
4. Perform predictions
5. Export final dataset

6. Open Power BI dashboard

Appendix E: GitHub Repository

The GitHub repository contains:

- Source code
- Datasets
- Reports
- Dashboard screenshots
- Presentation files

<https://github.com/Tgoswami1/Fake-Review-Detection-in-Education-Sector/tree/main>